Online Supplement to Hard to Beat: Historical Rules versus Conditional Quantile Models for Peak Net-Position Limits

This supplement retains the existing proofs and complete principal adverse-result inventories, with links to the unchanged longer archives. It accompanies the v10 manuscript and retains the accepted frozen simulation, same-dealer daily aggregation and fixed-parameter full-prefix backcast. Each addition was frozen before its own new output within a post-result / exploratory revision. The two original primary comparisons retain their values and roles. The earlier calibrated design and stability screen retain separate sections. The v10 additions report paired menu isolation, independently reported CFTC categories, complete model confidence sets and fixed limit-cost regions.

Tables format saved native fields. Loss percentages are the existing relative differences against the named reference; probabilities are labelled as such. Different source-reported monetary units are not pooled. NA and failed rows remain distinct from complete results.

The complete earlier supplement, full comparison families, all candidate tables, annual and sparse sensitivities, numerical details and source guides remain accessible through EXTENDED RESULTS. This index is an access route, not a substitute for compatible input data or independent validation.

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S1 Known-floor projection and strict quantile coverage

We forecast the sampled maximum of the absolute position, including the observed issue position. Write $B = |N_t|$ and let $Y \geq B$ almost surely, with B measurable at the forecast origin. For $0 < \tau < 1$, the pinball score is $\rho_\tau(u) = \tau u_+ + (1-\tau)(-u)_+$. A lower inverse conditional quantile is $q_\tau = \inf\{c : F(c \mid \mathcal{F}_t) \geq \tau\}$.

Proposition 1: floor projection. Replacing a report c by $\max(B, c)$ weakly reduces its score at every possible outcome. At the population conditional quantile, the strict breach probability is at most $1-\tau$. These are distinct statements: projection applies to any report; the tail qualification requires the true conditional quantile. Here $q = q_\tau$, and α denotes the stated nominal tail bound.

$$\rho_\tau(Y - c) - \rho_\tau(Y - \max\{B, c\}) = \tau(B - c)_+, \qquad \Pr(Y > q \mid \mathcal{F}_t) \leq \alpha.$$

Proof. If $c \geq B$, the replacement changes nothing. If $c < B$, both errors $Y - c$ and $Y - B$ are nonnegative, so their score difference is $\tau(B - c)$. This proves the pointwise identity without a density assumption. Right continuity of the conditional distribution gives $F(q_\tau \mid \mathcal{F}_t) \geq \tau$, hence $\Pr(Y > q_\tau \mid \mathcal{F}_t) \leq 1 - \tau$. An atom at the quantile can make this strict; exact nominal hit frequency is not required for a valid upper quantile. A fitted report does not automatically inherit this population claim. *QED*

Why the floor matters. The sampled maximum equals the current magnitude whenever the recorded future path remains inside its current absolute-position band. This can create an atom even when later signed observations are continuous. The existence of a floor alone does not imply an atom: an outcome uniformly distributed above the floor provides a counterexample. Equality-based coverage tests therefore diagnose their stated equalities, rather than every possible valid upper-quantile report.

Endpoint marginals do not identify the peak. The exact artificial constructions in the formal example record hold each future-time marginal fixed while changing temporal dependence. Comonotone and disjoint rare events then give different peak quantiles. The example is a logical construction, not another financial dataset or an estimated mechanism. Forecast evaluation must use the ordered path response and the information available at its actual origin.

Takeaway. Known-floor projection is an elementary pointwise improvement. It does not establish empirical calibration, identify the conditional distribution, or turn a fitted endpoint forecast into a forecast of a path maximum.

S2 Classical endpoint-to-peak bound

Proposition 2: the classical endpoint bound. Let future real increments $X_1, \ldots, X_H$ be independent and individually symmetric about zero. Set $S_0 = 0$, $S_h = \sum_{j=1}^{h} X_j$, and fix a starting position x, with absolute magnitude B. Equal increment laws and finite moments are unnecessary. For the population absolute-displacement quantile at level $1 - \alpha/2$, $q = \inf\{c : \Pr(|S_H| \leq c) \geq 1 - \alpha/2\}$, the classical scalar maximal inequality and the triangle inequality give

$$\Pr\left(\max_{0 \leq h \leq H} |x + S_h| > B + q\right) \leq \Pr\left(\max_h |S_h| > q\right) \leq 2\Pr(|S_H| > q) \leq \alpha.$$

This is a direct application of the classical Lévy inequality, not a new maximal inequality; see Sussman (2023), Proposition 3.4.

Proof, including strict events and atoms. For $c \geq 0$, let η be the first index with $|S_\eta| > c$. On $\{\eta = k, S_k > c\}$, the remaining sum $R_k = S_H - S_k$ is independent of the information through k and is symmetric. Therefore $\Pr(R_k \geq 0) \geq 1/2$, including a possible atom at zero, and that weak sign keeps $S_H > c$. On the corresponding negative crossing, $R_k \leq 0$ keeps $S_H < -c$, also with probability at least one half. Summing over the disjoint crossing events gives $\Pr(\max_h |S_h| > c) \leq 2\Pr(|S_H| > c)$. The argument covers $c = 0$. Finally, $\max_{0 \leq h \leq H} |x + S_h| \leq |x| + \max_h |S_h|$, and right continuity gives the required strict tail bound at q. *QED*

Conditional use and empirical implementation. For an observed random anchor, the same argument must hold under its forecast-origin conditional law, with the corresponding conditional population endpoint quantile. Conditional independence and zero symmetry cannot be inferred from unconditional marginals. The empirical endpoint plug-in adds the current floor to a historical absolute-displacement order statistic. Overlapping labels, estimated quantiles and changing dependent histories provide no automatic inheritance of this population bound or a pinball-risk ordering.

Two different failure mechanisms. Dependence can destroy the inequality: take $X_1 = Z$, $X_2 = -Z$, with $Z \in \{-1, 1\}$ equally likely. The endpoint is zero, while the sampled maximum is one. Separately, replacing the required absolute-displacement quantile by a generic one-sided endpoint quantile can fail even for independent symmetric increments. The exact finite example is preserved in the formal example record. These logical counterexamples are separate from the Gaussian Monte Carlo cases below; a nonviolating simulation does not prove a theorem extension.

Takeaway. The bound supplies a clear population benchmark under explicit assumptions; the empirical report remains a model to validate, not a coverage certificate.

S3 Monte Carlo mechanisms: original increment persistence

These are the existing fixed-seed Gaussian experiments, not newly run simulations or calibrated replicas of either financial source. Every reported rate is a probability, not a percentage. The peak includes the initial level; the endpoint event concerns absolute displacement from that initial level. Each path cell has 100000 paths. L/Stat starts the level AR in stationarity; L/Abs fixes its absolute-level tail anchor; L/Sig is the separate signed-level tail-anchor sensitivity. I/Zero accumulates stationary AR increments from a zero position. M denotes a stationary-marginal endpoint threshold; C denotes a distinct fixed-start conditional endpoint oracle. Threshold partners sharing a seed use the same paths, not independent repetitions.

Table S1: Original increment-autoregression simulation, unchanged. Serial persistence here concerns increments, not fitted position levels.

Horizon	Persistence ϕ	Process	Seed	Endpoint threshold	Peak breaches	Peak breach probability	Monte Carlo standard error	Endpoint breach probability
6	0.00	I/Zero	20267023	5.490	3035	0.03035	0.00054	0.02454
6	0.70	I/Zero	20267093	10.092	2597	0.02597	0.00050	0.02573
6	0.85	I/Zero	20267108	11.634	2550	0.02550	0.00050	0.02550
6	0.98	I/Zero	20267121	13.190	2557	0.02557	0.00050	0.02557
24	0.00	I/Zero	20285023	10.981	3819	0.03819	0.00061	0.02504
24	0.70	I/Zero	20285093	24.599	2772	0.02772	0.00052	0.02475
24	0.85	I/Zero	20285108	33.394	2619	0.02619	0.00051	0.02525
24	0.98	I/Zero	20285121	49.784	2572	0.02572	0.00050	0.02572

The independent-increment rows satisfy the classical premise. Positive increment coefficients violate independence without necessarily causing a numerical breach of the bound. Their favourable rates are not a general result for serial dependence.

S4 Mean-reverting levels and negative-increment sensitivities

The corrected level process is $N_h = \phi N_{h-1} + \varepsilon_h$, with independent Gaussian innovations of variance $1 - \phi^2$ and unit stationary level variance. Its stationary increments have lag correlation $\rho_X(k) = -(1 - \phi)\phi^{k-1}/2$, so this mechanism differs from a positively persistent increment AR. The fitted financial coefficient describes levels, making the level equation the relevant illustrative mechanism. The negative-increment family is a separate sensitivity, with stationary increments and nonstationary cumulative levels.

Table S2: All corrected primary cases. Mean-reverting level rates are conservative in these finite cases; the negative-increment result must also be retained.

Horizon	Persistence ϕ	Process	Seed	Endpoint threshold	Peak breaches	Peak breach probability	Monte Carlo standard error	Endpoint breach probability
6	0.68	L/Stat/M	20567123	3.009	175	0.00175	0.00013	0.02462
6	0.68	L/Abs/M	20567124	3.009	0	0.00000	0.00000	0.10447
6	0.85	L/Stat/M	20567133	2.502	300	0.00300	0.00017	0.02553
6	0.85	L/Abs/M	20567134	2.502	14	0.00014	0.00004	0.08227
6	0.98	L/Stat/M	20567143	1.071	913	0.00913	0.00030	0.02434
6	0.98	L/Abs/M	20567144	1.071	372	0.00372	0.00019	0.03623
6	-0.30	I/Zero/M	20567223	4.244	3563	0.03563	0.00059	0.02524
6	-0.60	I/Zero/M	20567233	3.127	4439	0.04439	0.00065	0.02558
24	0.68	L/Stat/M	20585123	3.170	636	0.00636	0.00025	0.02495
24	0.68	L/Abs/M	20585124	3.170	1	0.00001	0.00001	0.11416
24	0.85	L/Stat/M	20585133	3.138	402	0.00402	0.00020	0.02520
24	0.85	L/Abs/M	20585134	3.138	1	0.00001	0.00001	0.11159
24	0.98	L/Stat/M	20585143	1.965	768	0.00768	0.00028	0.02468
24	0.98	L/Abs/M	20585144	1.965	85	0.00085	0.00009	0.06201
24	-0.30	I/Zero/M	20585223	8.167	4281	0.04281	0.00064	0.02476
24	-0.60	I/Zero/M	20585233	5.701	5032	0.05032	0.00069	0.02516

All saved level cases have peak rates below the nominal threshold. The largest negative-increment point estimate occurs at the longer horizon and the more negative coefficient. Its rate is 0.05032, with a saved excess of 0.032 percentage points and Monte Carlo SE 0.00069. This is not demonstrated population failure. General dependence invalidates the classical premise regardless of the observed direction. Zero observed counts also do not establish zero population probability.

S5 All secondary simulation sensitivities

For stationary random starts, absolute endpoint displacement has variance $2(1-\phi^H)$. At a fixed start x, its conditional Gaussian mean and variance are $(\phi^H-1)x$ and $1-\phi^{2H}$. Thus holding the stationary-marginal endpoint threshold across starts does not condition on a large anchor. The conditional-oracle sensitivities below are distinct thresholds on shared paths; neither is an oracle for the maximum.

Table S3: All secondary anchor and conditional-threshold sensitivities. Conditional and marginal endpoint calibration remain distinct.

Horizon	Persistence ϕ	Process	Seed	Endpoint threshold	Peak breaches	Peak breach probability	Monte Carlo standard error	Endpoint breach probability
6	0.68	L/Abs/C	20567124	3.717	0	0.00000	0.00000	0.02411
6	0.68	L/Sig/M	20567125	3.009	2	0.00002	0.00001	0.06261
6	0.68	L/Sig/C	20567125	3.433	1	0.00001	0.00001	0.02476
6	0.85	L/Abs/C	20567134	3.036	0	0.00000	0.00000	0.02488
6	0.85	L/Sig/M	20567135	2.502	21	0.00021	0.00005	0.05689
6	0.85	L/Sig/C	20567135	2.840	4	0.00004	0.00002	0.02607
6	0.98	L/Abs/C	20567144	1.146	215	0.00215	0.00015	0.02503
6	0.98	L/Sig/M	20567145	1.071	464	0.00464	0.00021	0.03184
6	0.98	L/Sig/C	20567145	1.117	340	0.00340	0.00018	0.02521
24	0.68	L/Abs/C	20585124	3.920	0	0.00000	0.00000	0.02567
24	0.68	L/Sig/M	20585125	3.170	9	0.00009	0.00003	0.06486
24	0.68	L/Sig/C	20585125	3.605	1	0.00001	0.00001	0.02514
24	0.85	L/Abs/C	20585134	3.880	0	0.00000	0.00000	0.02495
24	0.85	L/Sig/M	20585135	3.138	3	0.00003	0.00002	0.06306
24	0.85	L/Sig/C	20585135	3.571	0	0.00000	0.00000	0.02484
24	0.98	L/Abs/C	20585144	2.298	20	0.00020	0.00004	0.02518
24	0.98	L/Sig/M	20585145	1.965	146	0.00146	0.00012	0.04607
24	0.98	L/Sig/C	20585145	2.179	56	0.00056	0.00007	0.02556

The complete correction contains 34 threshold rows on 22 initialized path samples. The retained CSV and portable readback provide exact endpoint moments, deviations from the nominal probability, and the single-prespecified-row bound for zero counts. None estimates financial innovation laws, changing fitted parameters or empirical-quantile uncertainty.

S6 Empirical endpoint plug-in: all original cells

The endpoint plug-in uses each completed historical episode's absolute endpoint displacement and adds its specified empirical quantile to the actual current-position floor. It shares the historical rule's completed origin pool and fixed refit schedule, with no fitted tuning parameter. The operational empirical rank and overlapping financial history are distinct from the known population quantile in the classical proposition. The report therefore has to earn its financial coverage and score performance empirically.

Table S4: Every original-cell endpoint plug-in comparison. A population bound does not imply a lower empirical pinball score.

Source	Horizon / history	Endpoint vs historical (%)	Endpoint vs regression (%)	Breaches endpoint / historical
FX	6/720	30.93	33.76	101/332
FX	6/2880	30.39	52.98	75/360
FX	24/720	10.85	-15.71	136/427
FX	24/2880	7.94	12.16	174/357
Bills	6/52	3.37	-43.62	10/11
Bills	6/260	-11.77	-41.49	15/28
Bills	24/52	26.32	-61.42	42/33
Bills	24/260	9.87	-47.64	22/12

Table S5: All separate endpoint-plug-in working-reference contrasts. Adverse score comparisons remain alongside lower-breach reports.

Source	Horizon / history	Reference	Loss-test statistic	Normal p-value	Small-sample p-value	Holm-adjusted p-value
FX	6/720	F	7.891	2.988e-15	3.658e-15	4.483e-14
FX	6/720	QR	6.608	3.908e-11	4.37e-11	5.081e-10
FX	6/2880	F	7.790	6.695e-15	8.123e-15	9.373e-14
FX	6/2880	QR	12.832	1.089e-37	3.451e-37	1.743e-36
FX	24/720	F	1.848	0.06462	0.06562	0.5169
FX	24/720	QR	-1.673	0.09434	0.09556	0.6604
FX	24/2880	F	2.058	0.03958	0.04032	0.3958
FX	24/2880	QR	1.530	0.1259	0.1273	0.7002
Bills	6/52	F	0.971	0.3317	0.3422	0.7002
Bills	6/52	QR	-2.778	0.005471	0.006887	0.06018
Bills	6/260	F	-1.345	0.1785	0.1884	0.7002
Bills	6/260	QR	-3.031	0.002441	0.003236	0.02929
Bills	24/52	F	1.569	0.1167	0.1522	0.7002
Bills	24/52	QR	-1.991	0.04645	0.06948	0.418
Bills	24/260	F	1.092	0.2746	0.3183	0.7002
Bills	24/260	QR	-1.516	0.1294	0.1664	0.7002

The family and its doubled-bandwidth sensitivities are separate from the original regression family. The complete tables keep every adverse direction; no negative comparison is removed by describing the population inequality as conservative. The linked records preserve source-reported mean error amounts and every numerical status.

S7 AR residual replay and its adverse comparisons

At each original refit, estimate an intercept-bearing level recursion on each unique mature adjacent transition once. Replaying a historical ordered residual block from the current position gives the exact algebraic identity

$$M_{i,h}(x) = N_{i+h} + \widehat{\phi}^{\,h}(x - N_i), \qquad M_{i,0}(x) = x.$$

The empirical peak uses the same completed scenario origins, response rank, floor and refit schedule as its matched historical controls. Finite negative or above-unit coefficients were eligible, not clipped; all observed fitted coefficients in the archived run happened to lie inside the unit interval. Their descriptive means are not identified data-generating parameters.

Table S6: Complete original autoregressive replay comparison. Lower scores can accompany more strict breaches.

Source	Horizon	History	Mean fitted $\widehat{\phi}$	Replay vs translated (%)	Replay vs regression (%)	Breaches replay / translated
FX	6	720	0.981	-1.54	2.45	282/255
FX	6	2880	0.985	-0.55	16.20	338/268
FX	24	720	0.981	2.39	-25.32	401/289
FX	24	2880	0.985	5.15	8.68	444/292
Bills	6	52	0.676	-13.44	-52.66	22/11
Bills	6	260	0.847	-18.89	-47.21	31/28
Bills	24	52	0.711	35.13	-59.43	56/33
Bills	24	260	0.850	-24.12	-63.86	59/12

The replay improves on original regression in every original Treasury cell but loses in most FX cells. Its comparisons with the translated-path rule have mixed directions, while strict breaches increase in every row above. Sparse-grid and annual signs can reverse. The full paired, annual and fixed-grid tables remain in the AR archive. These are post-result / exploratory same-data calculations, not independent validation.

S8 Ordinary quantile ranks and minimizer endpoints

The ordinary empirical quantile and the corrected empirical rule use the same completed pool, refit times and issue floor. The former takes the lower empirical pinball minimizer; the latter uses the specified add-one rank. At integer probability-times-pool-size values, adjacent reports can be the two endpoints of the same empirical minimizer interval. Calling the lower report a uniquely optimal comparator would therefore be wrong. At the short weekly pool, the noninteger rank distinction differs.

Table S7: Complete ordinary-rank ablation. Lower empirical reports increase breach occasions and need not improve score.

Source	Horizon	History	Regression vs empirical (%)	Empirical vs historical (%)	Breaches empirical / historical / regression
FX	6	720	-1.967	-0.153	337/332/531
FX	6	2880	-14.773	0.008	365/360/376
FX	24	720	31.508	-0.003	432/427/924
FX	24	2880	-3.783	0.026	359/357/531
Bills	6	52	87.086	-1.995	14/11/84
Bills	6	260	46.058	3.237	32/28/54
Bills	24	52	208.258	6.213	37/33/132
Bills	24	260	105.020	2.353	15/12/84

The reconstructed historical summaries match the original controls. The lower report increases strict hits and mean underprediction amount in every cell, but its loss direction is mixed. Every Treasury regression remains worse than the lower empirical report. This ablation isolates a rank convention on the same viewed record; it neither supplies a fresh validation sample nor establishes universal superiority of either finite-sample quantile endpoint. Exact counts, annual splits and fixed-grid rows are linked in the complete rank archive.

S9 Fixed-report error preferences

For a fixed report, changing the loss weight traces a line $R_a = aU + (1-a)V$, where $U = \text{mean}(P-C)_+$ and $V = \text{mean}(C-P)_+$. These are passive error amounts over all origins, not observed funding costs. Breach frequency averages an indicator and need not rank reports like either amount. The following source-reported monetary values are never pooled across FX and weekly aggregates.

Table S8: The four fixed headline comparisons. The longer-horizon FX result is sensitive to error weights.

Comparison	Under-prediction left	Under-prediction right	Unused allowance left	Unused allowance right	Equal-loss weight a
fx_h6_primary	156123.03	166531.06	4750560.18	5671390.59	none
fx_h6_matched	156123.03	185672.23	4750560.18	5526087.27	none
fx_h24_legacy	317678.12	149813.88	8658017.18	12121206.97	$\frac{86579744869851975}{90776350660766843}$
treasury_h6_primary	3131.86	1395.64	23688.09	28656.17	$\frac{24840389716843550}{33521504184965797}$

The equality column gives exact saved rational boundaries for the mixture weight defined above, not newly estimated costs. The short-horizon FX primary reduces both mean error amounts; its elementary-score curves can nevertheless cross because a Murphy diagram varies the elementary score threshold, a different comparison from reweighting these two unconditional amounts. The longer-horizon FX legacy pair reverses near the loss-weight ratio reported in the main text. Actual decision costs were not observed.

The complete frontiers below include every open region and every boundary tie. The full and sparse samples remain separate, and feasible reports outside the specified menu are not part of these menu-relative envelopes. All pairwise relations, dominated labels and exact arithmetic are additionally preserved in the preference archive.

S10 Complete preference-frontier source

The complete full-sample and fixed sparse-grid frontiers retain every boundary tie and open region. Their unchanged tables appear in the v9 Supplement, Tables S9–S10, and the exact rational source fields remain in the linked preference results. This supplement retains the error amounts and reversal above, while allocating the full new-method comparison tables below. No preference result was recomputed or removed from the repository.

S11 Complete weekly panel: definitions and first half

Each cell compares the same peak target, completed-label pool, origin sequence, fixed refit schedule and native monetary units. The original full regression, smaller predictor regression, expanded-ridge full regression and ordinary empirical report all remain visible. The additional series are related weekly aggregates, not independent individual-dealer replications. Changes in predictors and changes in the ridge menu are distinct interventions. Every listed comparison is post-result / exploratory.

The historical column gives mean pinball loss and strict hits. Other columns give saved percentage loss difference against that same historical rule and strict hits; negative percentages favour the column method. Original full regression, four-predictor regression and expanded-menu full regression are distinct fitted procedures (result codes QR33, QR4 and QR33+, respectively). I is the ordinary empirical-quantile report, with the endpoint qualification in Section S8. Across the archived panel, expanded full-regression median excess loss is 9.22 percent, compared with 92.63 percent for the original menu. Expanded regression still loses in 34 cells. This improvement does not erase the remaining failures or identify a causal feature effect.

Table S9: Complete panel, first half. Additional regularization leaves substantial adverse cases.

Series	Horizon / history	Historical loss / breaches	Original regression % / hits	Four-predictor regression % / hits	Expanded-menu regression % / hits	Empirical % / breaches
PDPOSMBS-TOT	6/52	921.7 / 21	170.97 / 101	3.24 / 27	5.82 / 28	3.26 / 27
PDPOSMBS-TOT	6/260	945.6 / 26	7.36 / 19	1.17 / 27	7.36 / 19	1.96 / 27
PDPOSMBS-TOT	24/52	1753.7 / 39	215.55 / 145	5.09 / 43	66.24 / 104	3.49 / 42
PDPOSMBS-TOT	24/260	1611.9 / 25	26.79 / 29	1.04 / 26	0.72 / 24	1.14 / 26
PDPOSCS-TOT	6/52	530.3 / 16	52.75 / 48	-5.79 / 23	-4.70 / 23	-6.28 / 23
PDPOSCS-TOT	6/260	399.7 / 6	1.90 / 15	-1.10 / 6	-0.95 / 8	-1.56 / 7
PDPOSCS-TOT	24/52	579.4 / 16	73.78 / 63	1.85 / 24	2.97 / 25	2.12 / 23
PDPOSCS-TOT	24/260	569.9 / 6	27.46 / 3	-3.38 / 6	-3.06 / 5	-4.68 / 6
PDPOSFGS-TOT	6/52	400.7 / 16	107.22 / 77	-4.15 / 24	-4.29 / 20	-4.98 / 18
PDPOSFGS-TOT	6/260	358.3 / 6	-1.08 / 14	4.88 / 9	-6.04 / 6	-1.54 / 6
PDPOSFGS-TOT	24/52	476.2 / 36	128.67 / 64	13.36 / 37	10.67 / 37	4.52 / 38
PDPOSFGS-TOT	24/260	354.0 / 9	30.02 / 21	72.24 / 26	-0.97 / 10	-1.50 / 9
PDPOSTIPS-L2	6/52	287.3 / 26	217.18 / 125	6.74 / 29	2.64 / 28	1.69 / 26
PDPOSTIPS-L2	6/260	269.3 / 24	66.28 / 45	1.78 / 25	66.28 / 45	1.63 / 25
PDPOSTIPS-L2	24/52	641.0 / 48	307.36 / 181	24.88 / 69	152.16 / 146	1.88 / 49
PDPOSTIPS-L2	24/260	532.2 / 47	226.91 / 147	11.41 / 59	226.91 / 147	1.84 / 51
PDPOSTIPS-G11	6/52	42.3 / 23	196.70 / 118	1.71 / 29	6.68 / 41	-1.13 / 26
PDPOSTIPS-G11	6/260	41.8 / 14	90.31 / 38	63.18 / 29	-1.52 / 13	-2.53 / 14
PDPOSTIPS-G11	24/52	80.8 / 61	218.69 / 123	135.59 / 103	64.63 / 78	-0.01 / 63
PDPOSTIPS-G11	24/260	51.2 / 9	98.59 / 33	37.18 / 30	98.59 / 33	-1.09 / 10
PDPOSTIPS-G2	6/52	181.3 / 16	208.61 / 96	8.58 / 21	35.47 / 41	3.33 / 19
PDPOSTIPS-G2	6/260	179.4 / 19	63.47 / 36	10.78 / 24	63.47 / 36	-0.29 / 19
PDPOSTIPS-G2	24/52	377.9 / 31	277.81 / 131	11.77 / 42	40.47 / 60	-3.63 / 33
PDPOSTIPS-G2	24/260	286.0 / 23	109.33 / 66	120.12 / 58	109.33 / 66	0.29 / 23

S12 Complete weekly panel: second half

The second half follows the identical convention. All original and expanded results remain unchanged. Cross-series dependence, aggregation offsets, native-unit differences and overlapping economic positions preclude treating these cells as independent replications. Forecasts are fitted separately; observations and losses are not pooled into a single financial population.

Table S10: Complete panel, second half. The enlarged penalty menu is not uniformly better than historical rules.

Series	Horizon / history	Historical loss / breaches	Original regression % / hits	Four-predictor regression % / hits	Expanded-menu regression % / hits	Empirical % / breaches
PDPOSTIPS-G6L11	6/52	133.6 / 9	59.30 / 73	-1.65 / 12	-2.52 / 13	-0.79 / 12
PDPOSTIPS-G6L11	6/260	114.9 / 4	-5.72 / 14	-1.11 / 4	-1.17 / 4	-1.00 / 4
PDPOSTIPS-G6L11	24/52	147.0 / 10	76.48 / 92	-0.97 / 16	-2.43 / 17	0.44 / 15
PDPOSTIPS-G6L11	24/260	133.8 / 2	-26.42 / 1	-1.30 / 2	-1.33 / 2	-0.55 / 2
PDPOSGSC-L2	6/52	1232.9 / 15	215.90 / 129	-3.85 / 23	215.90 / 129	-2.12 / 19
PDPOSGSC-L2	6/260	1247.4 / 15	64.40 / 40	9.04 / 20	2.20 / 20	0.61 / 17
PDPOSGSC-L2	24/52	2270.8 / 46	276.55 / 165	182.80 / 96	131.63 / 93	5.02 / 55

Table S10: Continued.

Series	Horizon / history	Historical loss / breaches	Original regression % / hits	Four-predictor regression % / hits	Expanded-menu regression % / hits	Empirical % / breaches
PDPOSGSC-L2	24/260	2918.7 / 33	41.60 / 55	-0.19 / 33	0.12 / 32	0.14 / 34
PDPOSGSC-G2L3	6/52	792.4 / 24	121.15 / 93	145.25 / 112	25.51 / 40	3.86 / 28
PDPOSGSC-G2L3	6/260	1004.0 / 28	21.31 / 44	22.62 / 59	21.31 / 44	0.23 / 30
PDPOSGSC-G2L3	24/52	2101.9 / 47	112.95 / 135	129.55 / 151	112.95 / 135	1.78 / 54
PDPOSGSC-G2L3	24/260	2677.0 / 55	11.60 / 64	39.73 / 91	-2.06 / 52	1.93 / 55
PDPOSGSC-G3L6	6/52	764.9 / 12	87.61 / 82	75.44 / 72	10.10 / 29	-4.25 / 15
PDPOSGSC-G3L6	6/260	699.1 / 20	0.79 / 15	1.91 / 21	0.79 / 15	2.27 / 22
PDPOSGSC-G3L6	24/52	977.4 / 38	544.34 / 137	299.12 / 120	19.44 / 47	9.23 / 45
PDPOSGSC-G3L6	24/260	931.0 / 20	15.52 / 37	0.13 / 20	-5.76 / 17	0.56 / 20
PDPOSGSC-G6L7	6/52	542.3 / 16	253.66 / 142	89.00 / 79	48.88 / 63	-1.24 / 24
PDPOSGSC-G6L7	6/260	564.9 / 20	31.61 / 42	71.05 / 49	31.61 / 42	0.43 / 21
PDPOSGSC-G6L7	24/52	993.4 / 43	264.80 / 217	264.77 / 215	264.80 / 217	11.08 / 47
PDPOSGSC-G6L7	24/260	699.0 / 22	146.41 / 77	142.57 / 67	146.41 / 77	0.98 / 23
PDPOSGSC-G7L11	6/52	505.7 / 25	175.62 / 83	15.72 / 34	86.69 / 55	14.85 / 34
PDPOSGSC-G7L11	6/260	506.2 / 25	8.34 / 31	-0.06 / 25	8.34 / 31	0.44 / 26
PDPOSGSC-G7L11	24/52	1653.9 / 36	94.95 / 105	62.40 / 91	-0.51 / 39	2.38 / 36
PDPOSGSC-G7L11	24/260	1002.9 / 32	229.96 / 94	202.78 / 99	229.96 / 94	8.62 / 35

The expanded-menu selection precedes its evaluation, as documented in the E5 archive. The original panel's entire candidate/status inventory remains in the original panel archive, and the fitted expanded-menu predictions are not substitutes for unavailable methods or independent dealer data.

S13 Complete added-panel endpoint-plug-in comparison

The original added-panel endpoint plug-in was evaluated on the same task grids as the historical and regression reports. Each entry gives the saved loss percentage against the historical rule, followed by strict hits for the endpoint plug-in and historical report. These are all related aggregate tasks, not independent replications or source-pooled monetary losses.

Table S11: Every additional-panel endpoint-plug-in result. Fewer breaches do not guarantee a better score.

Series	Horizon 6 history 52	Horizon 6 history 260	Horizon 24 history 52	Horizon 24 history 260
PDPOSMBS-TOT	7.17; 11/21	7.33; 5/26	-17.87; 27/39	-3.92; 22/25
PDPOSCS-TOT	18.28; 11/16	17.56; 2/6	-5.06; 11/16	19.63; 4/6
PDPOSFGS-TOT	-4.74; 13/16	16.77; 0/6	-3.72; 37/36	-4.06; 12/9
PDPOSTIPS-L2	-1.89; 12/26	-7.42; 19/24	-34.60; 30/48	-4.52; 46/47
PDPOSTIPS-G11	13.93; 3/23	27.71; 1/14	-16.39; 16/61	27.11; 0/9
PDPOSTIPS-G2	19.45; 4/16	23.00; 4/19	6.36; 27/31	2.09; 20/23
PDPOSTIPS-G6L11	9.67; 5/9	36.27; 0/4	5.03; 4/10	12.63; 0/2
PDPOSGSC-L2	20.65; 6/15	-2.64; 6/15	0.93; 27/46	-22.23; 24/33
PDPOSGSC-G2L3	17.31; 15/24	-14.81; 12/28	22.35; 49/47	-9.35; 40/55
PDPOSGSC-G3L6	20.57; 9/12	3.66; 8/20	-11.37; 20/38	-5.73; 17/20
PDPOSGSC-G6L7	0.95; 12/16	-10.15; 15/20	1.69; 49/43	5.19; 21/22
PDPOSGSC-G7L11	4.92; 11/25	1.05; 12/25	-20.33; 26/36	1.88; 26/32

The complete original panel CSV also preserves every original regression candidate, selection result, native-unit mean error amount and failure status. This table adds no refit or parameter selection and does not extend the independent-increment population guarantee to these financial series.

S14 Earlier-period fixed selection rules

The earlier-period experiment applies the saved fixed selection rules to familiar historical data. These origins already have validation and evaluation roles in the current archive, including actual rolling-fold scoring. They cannot now be treated as untouched evidence for the expanded backcast. The experiment remains retrospective, not external or prospective validation. R0 denotes the fixed minimum-validation- loss rule. R2 is the predesignated main one-standard-error selection rule, not a result-dependent replacement chosen for its subsequent performance. The rule design precedes its calculations but follows earlier research results.

R1 adds the ordinary empirical quantile to the finite regression candidates. R2 compares each candidate's validation losses with the same best-mean candidate, using the paired-difference Bartlett-HAC standard error. Candidates within that tolerance qualify, and the fixed rule prefers the strongest shrinkage, including the empirical report. This paired HAC tolerance is a time-series adaptation of a one-standard-error preference, not the textbook formula, a significance test or a noninferiority guarantee. It does not account for the whole uncertainty from selecting the best candidate on that segment.

R3 applies the same preference to prespecified earlier rolling folds and the outer validation segment, with origin-equal weighting and maturity gaps. Its HAC pairs use actual source-index distances rather than compressing gaps. Without an eligible earlier fold, the saved design makes it identical to R2; this is not a numerical fallback. Earlier-period results were sealed before the already-inspected-period calculations that could reuse those origins.

Table S12: Every earlier-period minimum-loss and one-standard-error comparison. The one-standard-error rule misses the recorded success conditions.

Series	Horizon / history	Minimum-loss rule (%)	One-standard-error rule (%)	Breaches minimum-loss / one-standard-error	Historical breaches
PDPOSGS-B	6/52	11.22	7.43	10/10	7
PDPOSGS-B	24/52	181.39	181.39	27/27	8
PDPOSMBS-TOT	6/52	-4.11	3.83	7/6	4
PDPOSMBS-TOT	24/52	48.55	26.55	14/13	10
PDPOSCS-TOT	6/52	-11.56	-11.56	0/0	0
PDPOSCS-TOT	24/52	-4.12	1.02	4/4	2
PDPOSFGS-TOT	6/52	-27.32	-27.32	6/6	0
PDPOSFGS-TOT	24/52	-3.87	-3.60	4/3	2
PDPOSTIPS-L2	6/52	-7.06	-7.06	2/2	1
PDPOSTIPS-L2	24/52	-23.33	-1.55	4/1	1
PDPOSTIPS-G11	6/52	-5.55	-5.79	5/5	4
PDPOSTIPS-G11	24/52	5.67	-0.87	15/15	15
PDPOSTIPS-G2	6/52	15.16	15.16	8/8	0
PDPOSTIPS-G2	24/52	30.27	30.27	14/14	10
PDPOSTIPS-G6L11	6/52	283.88	28.58	17/6	0
PDPOSTIPS-G6L11	24/52	-4.03	-2.70	2/1	0
PDPOSGSC-L2	6/52	48.04	16.09	17/15	14
PDPOSGSC-L2	24/52	40.77	40.77	34/34	30
PDPOSGSC-G2L3	6/52	1.40	11.14	12/13	8
PDPOSGSC-G2L3	24/52	-11.22	15.48	15/20	15
PDPOSGSC-G3L6	6/52	2.63	2.35	11/11	10
PDPOSGSC-G3L6	24/52	15.62	6.92	29/28	26
PDPOSGSC-G6L7	6/52	19.03	19.03	6/6	1
PDPOSGSC-G6L7	24/52	2.05	1.82	20/19	19
PDPOSGSC-G7L11	6/52	19.18	12.77	13/13	10
PDPOSGSC-G7L11	24/52	8.48	2.70	29/29	27
fx_hourly	6/720	2.44	0.62	79/73	63
fx_hourly	6/2880	-16.62	-6.28	20/20	16
fx_hourly	24/720	-4.36	-4.36	47/47	32
fx_hourly	24/2880	-1.42	-1.40	1/1	0

All recorded success criteria are NOT_MET, NOT_MET and NOT_MET, respectively. R2's worst weekly excess is 181.39 percent, and its weekly median is 5.37 percent. The long-history FX gain also fails the prewritten retention criterion. This adverse result remains a robustness finding rather than a contribution claim. Complete inspected-period and all-rule rows remain linked in the R6 archive.

S15 Selection failures and replication boundaries

Required validation-candidate failure blocks the dependent rule; it is neither retried nor replaced by a shorter mean. The complete noncomplete report inventory is below. Every listed row has the saved reason `REQUIRED_VALIDATION_CANDIDATE_FAILED`. Those failures remain part of the evidence even though other rules in the same task can complete.

Table S13: All noncomplete fixed-rule report rows. Failed mandatory candidates remain failures.

Phase	Series	Horizon / history	Rule	Status
retro	fx_hourly	6/720	R3	FAILED

Table S13: Continued.

Phase	Series	Horizon / history	Rule	Status
retro	fx_hourly	24/720	R3	FAILED
inspected	fx_hourly	6/720	R3	FAILED
inspected	fx_hourly	24/720	R3	FAILED

The Treasury earlier-period structural comparison uses a shorter history than the original primary. It is explicitly an analogue, not an exact earlier replication of the long-history primary. The source-reported comparison and its definition are preserved in the primary-structure table. All original primary values remain unchanged. Reclassifying the analogue as an exact primary replication, or calling familiar retrospective FX data an unseen dealer, would change the scientific claim without adding evidence.

S16 Finite delayed calibration and headroom SAV

The finite delayed ACI-style wrapper adapts past strict hits after their entire peak labels mature. Its score pool contains only completed residual scores and uses the specified finite memory. The base-plus-score quantile is projected onto the known floor. The adaptive alpha state is projected to the current feasible range; the rank-zero convention issues the floor. Gamma was selected by coverage distance, then pinball loss and the fixed tie rule, a coverage-first selection criterion distinct from original regression fitting. No canonical ACI theorem is imported to this modified procedure.

The headroom SAV method adapts a CAViaR recursion, rather than replicating a return-VaR model. The completed headroom response is normalized by its training-only scale; the recursive report is multiplied back by that same scale and then floor-projected. Its fixed bounded optimizer and starts retain all eligibility and convergence failures; a complete fit is not a proof of a global optimum. Selection state carries across the unissued chronological gap.

Table S14: Complete delayed-wrapper and headroom-recursion inventory. Two long-horizon wrapper cells remain unavailable.

Source	Horizon / history	Method	Mean loss	Breaches	Status
FX	6/720	ACI_F	472677.77	331	COMPLETE
FX	6/720	ACI_QR	439672.91	352	COMPLETE
FX	6/720	SAV	449237.37	349	COMPLETE
FX	6/2880	ACI_F	463271.47	328	COMPLETE
FX	6/2880	ACI_QR	393795.06	329	COMPLETE
FX	6/2880	SAV	456696.22	364	COMPLETE
FX	24/720	ACI_F	921402.08	322	COMPLETE
FX	24/720	ACI_QR	1020220.09	393	COMPLETE
FX	24/720	SAV	781586.36	439	COMPLETE
FX	24/2880	ACI_F	791441.18	323	COMPLETE
FX	24/2880	ACI_QR	755982.70	294	COMPLETE
FX	24/2880	SAV	756811.00	333	COMPLETE
Bills	6/52	ACI_F	3818.40	18	COMPLETE
Bills	6/52	ACI_QR	3489.82	29	COMPLETE
Bills	6/52	SAV	2974.32	16	COMPLETE
Bills	6/260	ACI_F	2792.91	7	COMPLETE
Bills	6/260	ACI_QR	3013.16	6	COMPLETE
Bills	6/260	SAV	2892.23	29	COMPLETE
Bills	24/52	ACI_F	4368.10	38	COMPLETE
Bills	24/52	ACI_QR	6250.53	47	COMPLETE
Bills	24/52	SAV	4182.40	35	COMPLETE
Bills	24/260	ACI_F	NA	NA	NA
Bills	24/260	ACI_QR	NA	NA	NA
Bills	24/260	SAV	3100.97	25	COMPLETE

The NA wrapper rows have the saved reason `VALIDATION_INELIGIBLE`; no test is substituted for the absent forecasts. All losses are in each source's native units and must not be pooled.

S17 Wrapper comparisons, uncertainty and missing cells

Comparing wrapped regression with wrapped historical reports is a different question from comparing either wrapper with its unwrapped base. The following table retains every within-cell comparison. A lower coverage

distance measures closeness of empirical hits to the target probability, not conditional coverage or lower monetary loss.

Table S15: Every paired wrapper comparison. Calibration adjustment preserves the long-history short-horizon FX score gain but does not produce universal dominance.

Source	Horizon / history	Regression vs historical (%)	Breaches regression / historical	Regression adjustment	Historical adjustment
FX	6/720	-6.98	352/331	0.00313	0.00004
FX	6/2880	-15.00	329/328	0.00034	0.00049
FX	24/720	10.72	393/322	0.00932	0.00140
FX	24/2880	-4.48	294/323	0.00562	0.00125
Bills	6/52	-8.61	29/18	0.05469	0.01498
Bills	6/260	7.89	6/7	0.02834	0.02473
Bills	24/52	43.09	47/38	0.11968	0.08718
Bills	24/260	NA	NA	NA	NA

The working-reference comparison inventory contains 56 fixed slots, with 5 raw p values unavailable. The reported Holm bookkeeping keeps those slots; mapping an NA component to a bookkeeping value does not create an observed p value or validate failed reference assumptions. HLN references remain separately identified. Exact update equations, gamma candidates, selection freezes, numerical receipts and all comparisons remain in the v6 replay guide and the complete WP2 inventory. The original primaries are not overwritten by these post-result / exploratory extensions. The upper-horizon NA rows and adverse SAV comparisons remain visible rather than becoming a favourable subset.

S18 Untuned forecast combinations

Fixed report combinations use the matched historical and original regression reports at identical origins. Their arithmetic mean is untuned. The median also requires the original selected plain local-quantile report; neither a smaller predictor model nor the endpoint plug-in substitutes for that missing constituent. Combining floor-respecting reports preserves the floor, but does not guarantee a conditional quantile or a score below the best constituent. Identical reports generate an NA equal-loss p value rather than a manufactured significance result.

Table S16: All original-cell combination comparisons. A report combination can improve one constituent while losing to the other.

Source	Horizon / history	Rule	vs historical (%)	vs regression (%)	Breaches
FX	6/720	Mean	-6.91	-4.90	338
FX	6/720	Median	-4.81	-2.75	325
FX	6/2880	Mean	-10.61	4.87	330
FX	6/2880	Median	-11.17	4.22	327
FX	24/720	Mean	5.50	-19.77	533
FX	24/720	Median	5.00	-20.15	514
FX	24/2880	Mean	-8.94	-5.39	345
FX	24/2880	Median	-3.74	0.01	416
Bills	6/52	Mean	-3.15	-47.18	26
Bills	6/52	Median	0.00	-45.46	11
Bills	6/260	Mean	2.77	-31.84	40
Bills	6/260	Median	-3.98	-36.32	28
Bills	24/52	Mean	18.87	-63.69	54
Bills	24/52	Median	0.00	-69.46	33
Bills	24/260	Mean	-5.32	-54.88	28
Bills	24/260	Median	-11.51	-57.83	14

Table S17: Complete combination counts including the added panel. Missing median constituents remain NA.

Rule	Total	Complete	Unavailable	Lower loss than both	Lower loss than historical	Lower loss than regression	Equal loss to historical
Mean	56	56	0	14	16	54	0
Median	56	8	48	3	5	6	2

The added panel lacks the specified plain local-quantile constituent, so all its median rows remain unavailable. Mean combinations are complete, but lower loss than regression alone does not imply lower loss than the historical rule. All panel-specific values and every comparison remain in the complete combination record.

S19 Combination Murphy diagrams and comparison families

Elementary-score comparisons vary the score threshold while keeping the quantile level fixed. Exact integer-event signs at every unique breakpoint determine sample crossing before display interpolation. A picture is not used to select favourable thresholds. These matched combination contexts are distinct from the original FX primary with its shorter rolling reference.

Table S18: Every combination Murphy comparison in the original-primary contexts. Crossing prevents threshold-uniform sample dominance.

Source	Left-hand method	Right-hand method	Ranking	Positive breakpoints	Negative breakpoints
FX	MEAN_F_QR	F	CROSSING	217	15628
FX	MEAN_F_QR	QR	CROSSING	13562	2343
FX	MEDIAN_F_QR_KNN	F	CROSSING	909	13572
FX	MEDIAN_F_QR_KNN	QR	CROSSING	10827	2575
Bills	MEAN_F_QR	F	CROSSING	336	291
Bills	MEAN_F_QR	QR	CROSSING	179	454
Bills	MEDIAN_F_QR_KNN	F	CROSSING	262	274
Bills	MEDIAN_F_QR_KNN	QR	CROSSING	147	446

The complete normal-DM comparison family has 224 slots, of which 98 have NA raw p values. The family adjustment is bookkeeping for those saved working references, not error control over the complete historical analysis search. The inspected record, overlapping labels, repeated fitting and related aggregate series remain common to the methods. All original and adverse rows remain in the combination findings. Exact breakpoint signs support only the stated finite-sample score comparison; they do not certify general forecast dominance, optimal funding, or a causal benefit from acting on the limits.

S20 Training-defined position-change states

The states are low, middle and high issue-known RMS position-change states, not price volatility or yield regimes. Cutpoints come only from the training history. Reports and original primary means stay unchanged; the diagnostic divides the already viewed evaluation record without estimating a switching policy. Source-native group differences are not pooled.

Table S19: All three state groups for each original primary. The FX advantage remains negative in every group, while the weekly evidence remains adverse.

Source	State	Issues	Regression minus reference loss	Dependence-adjusted standard error
FX	low	1593	-58761.27	16492.94
FX	middle	1816	-27839.43	16257.70
FX	high	3216	-70387.91	17895.89
Bills	low	9	6696.66	1705.21
Bills	middle	25	5230.59	2608.15
Bills	high	243	810.88	466.48

Table S20: Complete state diagnostic reference family. A joint rejection need not identify a useful state rule.

Source	Diagnostic	Unadjusted p-value	Holm-adjusted p-value	Status
FX	joint_GW	1.46298e-06	5.8519e-06	OK
FX	group_heterogeneity	0.159383	0.159383	OK
Bills	joint_GW	0.0484878	0.0969756	OK
Bills	group_heterogeneity	0.000426772	0.00128032	OK

These finite-instrument and group-mean HAC references require unverified moment, dependence and covariance assumptions. A joint conditional- moment rejection can be driven by an unconditional mean difference; it is not evidence that a state-dependent trading or limit-selection policy works. The source-level reconstruction and all

cutpoints remain in the complete state record.

S21 Refitted target-level sensitivities

Changing the target quantile requires fitting the specified target- level procedure; it is different from reweighting the errors of a fixed report. The following saved sensitivities keep the original horizon and reference structure. The weekly primary and matched reference coincide and count once. All unavailable upper-target rows remain NA with `VALIDATION_INELIGIBLE`, rather than being replaced by a lower target or a favourable subset. Absolute loss values across different quantile levels measure different objectives.

Table S21: Complete target-level sensitivity. The lower-target directions recur, but upper-target evidence remains unavailable.

Source	Target τ	Reference	Mean loss difference	Regression breaches	Reference breaches	Status
FX	0.90	primary	-60715.44	686	620	OK
FX	0.90	matched	-67628.17	686	667	OK
Bills	0.90	primary	1175.03	73	55	OK
FX	0.99	primary	NA	NA	NA	NA
FX	0.99	matched	NA	NA	NA	NA
Bills	0.99	primary	NA	NA	NA	NA

The complete family contains 6 slots, with 3 raw p values unavailable. The fixed target-specific selections precede their evaluation; full settings and results remain in the target-level inventory. The saved lower-target directions do not establish robustness to an arbitrary tail level, actual shortage costs, or a new economic decision rule. Failed eligibility is a result of the specified procedure and remains part of the paper's boundary.

S22 Horizon coherence and late conditional-hit evidence

Realized peaks are ordered as the forecast horizon grows, but independently fitted reports use horizon-specific mature peak responses and can cross. The following exact strict comparisons keep all common issue origins. Repairing a long-horizon report upward can worsen pinball loss; target ordering alone does not identify a universally beneficial repair.

Table S22: Every original quantile-regression horizon-coherence comparison. Ordered targets do not force ordered fitted reports.

Source	History	Issues	Forecast crossings	Target violations
FX	720	6625	640	0
FX	2880	6625	140	0
Bills	52	277	91	0
Bills	260	277	80	0

The late fixed-terminal conditional-hit diagnostic uses the null $E[I_t \mid \mathcal{F}_t] \leq \alpha$, where $I_t = 1\{P_t > C_t\}$. Within each globally anchored maturity phase, a fixed upper-probability alternative supplies hit/no-hit factors with conditional expectation at most one under the stated null. The procedure mixes whole products over its fixed prior, then averages dependent phases. It does not multiply phases or select their maximum. The fixed-family threshold does not restore retrospective error control after viewed-sample diagnostic selection.

Table S23: Every threshold-crossing cell in the complete late diagnostic. A noncrossing elsewhere is not a calibration certificate.

Source	Horizon / history	Breaches / issues	Saved display value
FX	24/720	924/6625	8.499112682788E+7
Bills	6/52	84/277	1.310958812771E+8
Bills	6/260	54/277	2.204375927802E+3
Bills	24/52	132/277	3.814063640027E+4

The complete summary and phase tables, exact rational identities, all noncrossings and the interpretation boundaries remain in the e-value archive. This is a fixed-terminal diagnostic, not an anytime p value, optional-stopping guarantee, new estimator, or independent validation. Score, breach amount, frequency and conditional-hit evidence answer different questions.

S23 Numerical implementation and inference conventions

Completed labels and update clocks. At a scheduled refit κ, the latest mature peak response origin is $\kappa-H$. A history of W mature peak responses therefore uses $\{\kappa-H-W+1,\ldots,\kappa-H\}$, with every target ending by that refit. Training transforms, scales, coefficients and historical pools are held until the next scheduled refit. The current observed position may still change at each forecast issue. Replacing that issue anchor by the refit anchor changes the forecasting procedure.

Regression and local quantiles. The regression objective uses a training-normalized headroom response, an unpenalized intercept and a ridge penalty on slopes. Projection is applied when issuing the report and evaluating its validation score, not inside the fitted training objective. Local methods use training-only standardized predictors, fixed neighbour menus and duplicate-preserving order statistics. A selected whole-pool local quantile is an exact historical identity, not a failed-fit fallback. Scale-adjusted neighbours change responses as well as retrieval.

Numerical eligibility. The accepted solver records primal and dual residuals, objectives, feasibility and their gap. Ill-conditioned, nonfinite, unsupported-rank and failed candidate states retain their original statuses under each named selection contract. A mandatory-candidate failure blocks a procedure when its contract requires every candidate. The original full regression has that requirement; the four-feature selector may choose among completed eligible candidates. A failed selected evaluation never receives a replacement penalty or a shortened score. Changing the ridge menu changes the procedure even when feature definitions stay fixed. A numerical certificate does not establish financial accuracy, economic relevance, inferential validity or production continuity. No production continuity or fallback policy is evaluated here.

Working-reference inference. Saved unconditional loss differences use the recorded Bartlett-HAC convention; HLN and doubled-bandwidth sensitivities are labelled separately. Separate families and later reporting partitions do not erase the wider selection history. Coverage references distinguish full-grid Kupiec from the transition-conditional Christoffersen joint likelihood, which conditions on the initial hit. Small sparse grids and boundary likelihoods retain their original status. Neither a nonrejection nor a small working-reference p value proves conditional coverage or supplies selection-adjusted confirmation.

Common-pool construction distances. On one nonempty historical episode set, compare the historical headroom, translated-path and historical-peak reports using the same available order statistic. Each episode peak includes its starting observation. A translated path changes each level by the same anchor difference, so its absolute peak changes by at most that difference. Flooring is nonexpansive and commutes with the fixed order statistic. The headroom construction has the corresponding bound in the difference of anchor magnitudes. Thus

$$|C^{\text{headroom}}-C^{\text{peak}}|\le\max_i\big||N_i|-B\big|,\qquad |C^{\text{translated}}-C^{\text{peak}}|\le\max_i|N_i-x|.$$

For a proof of the order-statistic step, paired coordinatewise differences bounded by a radius imply that each empirical quantile lies below the other plus that radius; reverse the roles for the second inequality. These deterministic same-pool bounds do not order financial accuracy or coverage. Changing historical rows on one side invalidates the common-pool argument.

Exact menus, numerical tolerances, failures and diagnostic tables remain in EXTENDED RESULTS and the dated source archives.

S24 Frozen controlled comparison of inherited procedures

The simulation compares inherited reports for mean-reverting position levels, not positively autocorrelated increments and not a fitted reproduction of either financial source. It uses

$$N_t=\phi N_{t-1}+\varepsilon_t,\qquad \varepsilon_t=\sqrt{v_t}z_t,\qquad v_t=\omega+a\varepsilon_{t-1}^2+bv_{t-1},\qquad z_t\overset{\text{iid}}{\sim}\mathcal{N}(0,1).$$

The level persistences are $\phi\in\{0.7,0.9,0.98\}$, variance persistences $p=a+b\in\{0,0.9,0.98\}$, histories $W\in\{52,260,720,2880\}$, and horizons $H\in\{6,24\}$. For the increasing variance-persistence settings, the (ω,a,b) triples are $(1,0,0)$, $(0.1,0.05,0.85)$, $(0.02,0.05,0.93)$. Each innovation has unit unconditional variance under the specified recursion. Initial level and variance are fixed; a burn-in of 20000 observations precedes 3264 retained observations. This is a long-burn approximation, not an exact stationary initialization. The common hourly calendar has no simulated calendar effect.

There are 32 independent repetition streams, with master entropy 2026102408 and common normal innovations across conditions within a repetition. Validation uses origins 3072–3143; evaluation uses 3168–3239, with 72 overlapping issues and a refit every 24 records. Repetition, not issue or grid cell, is the Monte Carlo uncertainty unit. Training uses exactly W origins $r-H-W+1,\ldots,r-H$, with every label ending by refit r. The

current-position floor and query features use the actual issue state. The full model keeps its original predictors and the parsimonious model its original four; their penalty menus and failure rules differ, so their contrast is not an isolated intervention on predictor count.

For each cell, let A_j and B_j be regression and matched-historical mean loss in repetition j, and R the planned number of repetitions. The reported relative change and its delta-method Monte Carlo standard error are

$$\widehat{\Delta} = 100\left(\frac{\overline{A}}{\overline{B}} - 1\right), \qquad \mathrm{MCSE}(\widehat{\Delta}) = \frac{100\, s(A_j - \widehat{r}B_j)}{\sqrt{R}\,\overline{B}}, \qquad \widehat{r} = \frac{\overline{A}}{\overline{B}}.$$

Sample standard deviations use divisor $R-1$. Breach-rate errors use the between-repetition standard deviation divided by $\sqrt{R}$. Strict wins require $A_j < B_j$, with error $\sqrt{\widehat{\pi}(1-\widehat{\pi})/R}$. At zero or all wins that plug-in error is zero without implying certainty; the worst-case error bound is $0.5/\sqrt{R}$, or 8.84 percentage points here. The short evaluation segments, common innovations across cells, and fixed process family limit a regime claim.

The frozen descriptive clear-gain screen requires all 32 paired repetitions, $\widehat{\Delta} + 2\,\mathrm{MCSE}(\widehat{\Delta}) < 0$, and a strict-win fraction at least 0.75. Its target is $H = 6$, $W \geq 720$, $p \geq 0.9$. Support additionally requires at least 6 of 12 clear-gain target cells and none among the 60 outside cells. Each regression has only 2 target gains and 6 outside gains. The recorded conclusion is **MISMATCH**: both the target count and the no-outside requirement fail. This screen has no family-wide significance guarantee and supplies no new population endpoint-bound guarantee under dependent financial paths.

The complete two-regression 72-cell tables remain unchanged in v9 Tables S26–S27 and their native fields in the saved controlled simulation. All adverse, failed and unavailable statuses retain their original meanings. The preceding design and prediction outcomes refer to that entire grid, not a successful subset.

Candidate and procedure statuses. The complete inventory contains 6912 complete method tasks, 18431 complete validation candidates and 1 failed candidate. The failure is the four-feature candidate at zero-based repetition 23, $\phi = 0.9$, $p = 0$, $W = 52$, $H = 24$, and penalty 0.1. Its duality-gap criterion fails at 10000 iterations. The frozen eligible-completed-candidate rule selects penalty 0.01, whose full evaluation completes; this is not a retry or substitution of a failed selected evaluation. The original full-regression selector instead requires every declared validation candidate. Incomplete procedures retain NA full-procedure performance, with any available-pair summary labelled conditional on completion. The saved source summaries retain all cells and their full precision in `results/revision-v8/wp3/summary.json`; all six loss, breach and win heatmaps remain alongside them. Both complete numerical grids retain every cell.

S25 Same-dealer completed-day aggregation

The pinned hourly FX snapshot produces 930 nonempty source-naive calendar days under the frozen last-observation rule, rather than the approximate sample size in the planning instruction. Each report is issued only at the end of its completed source-naive day; the selected row's original timestamp is not treated as an earlier time at which the identity of the last row was already known. Missing days stay absent, and partial boundary days remain. The timezone and historical first-release availability are unknown. This is a last-record proxy, not a verified financial close or a peak over intraday excursions.

Histories and horizons count subsequent nonempty daily observations. The split copies the Treasury observation-index geometry, not its absolute calendar dates or a proportional split: validation runs from 2015-02-24 to 2015-04-24 and evaluation from 2015-05-24 to 2016-12-27 on 501 common issues. Raw or resampled positions and per-issue trajectories are not redistributed.

Table S24: Same-dealer daily sampled-position results; history and horizon count recorded nonempty calendar days. All methods share the declared evaluation grid; loss is scaled to USD millions for display only. The isolated parsimonious score gain has more breaches than its historical reference.

History	Horizon	Procedure	Penalty	Mean loss (USD m)	Loss change vs historical (%)	Breaches count / rate (%)
52	6	Historical headroom	–	1.049	–	36 / 7.186
52	6	33-feature regression	0.1	2.032	93.586	146 / 29.142
52	6	Four-feature regression	0.001	1.974	88.128	129 / 25.749
52	24	Historical headroom	–	1.553	–	66 / 13.174
52	24	33-feature regression	0.01	3.456	122.589	206 / 41.118
52	24	Four-feature regression	0.01	3.163	103.719	225 / 44.910
260	6	Historical headroom	–	0.964	–	30 / 5.988
260	6	33-feature regression	0.1	1.175	21.873	54 / 10.778
260	6	Four-feature regression	0.1	0.935	-2.974	37 / 7.385

Table S24: Continued.

History	Horizon	Procedure	Penalty	Mean loss (USD m)	Loss change vs historical (%)	Breaches count / rate (%)
260	24	Historical headroom	–	1.882	–	24 / 4.790
260	24	33-feature regression	0.001	2.642	40.335	99 / 19.760
260	24	Four-feature regression	0.001	2.177	15.627	46 / 9.182

Historical headroom wins seven of the eight named comparisons, but the four-feature longer-history, shorter-horizon daily exception lowers loss by about 3.0% with 37 regression breaches versus 30 historical breaches. Both regressions meet the fixed majority-based descriptive classification, not a dominance or coverage theorem. Same recorded source identity is held fixed while physical history, horizon, update cadence, calendar predictors, sampled response and evaluation span change jointly. The contrast cannot identify a pure frequency effect or supply independent dealer validation.

S26 Fixed-parameter full-prefix backcast and overlapping windows

The additional retrospective backcast uses the complete eligible prefix 3071–10438, with 7368 common issue origins before the original later validation. It retains the original later-selected parameters: full regression: $H = 6, \lambda = 0.01$, $H = 24, \lambda = 0.001$; four-feature regression: $H = 6, \lambda = 0.01$, $H = 24, \lambda = 0.01$. Coefficients, transformations and training labels obey $i + H \leq r$ at each refit, but those fixed penalties were chosen after the historical issue dates. Thus this is a fixed-parameter pseudo-out-of-sample backcast with later-selected parameters, not a real-time historical validation.

The full prefix contains all 1440 earlier reported backcast origins, with inclusive issue indices 8999–10438. Actual earlier rolling-validation targets at the longer horizon cover 3071–10462, exactly the new prefix's longer-horizon target range. Apparent issue gaps between old validation blocks are not untouched target paths. Three displayed overlapping/inspected windows therefore do not create a fresh independent third validation period. This **MISMATCH** remains regardless of the direction of the added score comparisons. No window effects, p-values or success counts are pooled.

Table S25: Six-step original regression across three inspected windows; the full-prefix window contains the earlier reported backcast, so this is not independent validation. Loss is scaled to USD millions for display; raw normal-HAC p-values are descriptive. Missing archived breach counts remain NA.

Window	Reference	Mean loss Reg. / hist. (USD m)	Change (%)	Breaches Reg. / hist.	DM	Raw p	Issues
Full prefix	Matched F2880	0.587 / 0.668	-12.167	416 / 282	-3.518	<0.001	7368
Full prefix	Rolling F720	0.587 / 0.619	-5.296	416 / 387	-1.918	0.055	7368
Prior backcast	Matched F2880	0.287 / 0.344	-16.624	20 / 16	-4.178	<0.001	1440
Prior backcast	Rolling F720	0.287 / 0.278	3.168	20 / NA	0.563	0.574	1440
Original	Matched F2880	0.386 / 0.453	-14.767	376 / 360	-5.866	<0.001	6625
Original	Rolling F720	0.386 / 0.442	-12.660	376 / 332	-5.182	<0.001	6625

Table S26: Complete full-prefix backcast comparisons; the two regressions use the original later-selected parameters. All planned comparisons are shown; raw normal-HAC p-values are descriptive. The window is already inspected and overlaps the earlier reported backcast.

Horizon	Regression	Reference	Mean loss Reg. / hist. (USD m)	Change (%)	Breaches Reg. / hist.	DM (raw p)
6	33-feature	Matched F2880	0.587 / 0.668	-12.167	416 / 282	-3.518 (<0.001)
6	33-feature	Rolling F720	0.587 / 0.619	-5.296	416 / 387	-1.918 (0.055)
6	Four-feature	Matched F2880	0.617 / 0.668	-7.620	290 / 282	-2.498 (0.012)
6	Four-feature	Rolling F720	0.617 / 0.619	-0.393	290 / 387	-0.177 (0.859)
24	33-feature	Matched F2880	1.470 / 1.783	-17.538	470 / 275	-1.997 (0.046)
24	33-feature	Rolling F720	1.470 / 1.431	2.730	470 / 477	0.315 (0.752)
24	Four-feature	Matched F2880	1.408 / 1.783	-21.027	355 / 275	-2.778 (0.005)
24	Four-feature	Rolling F720	1.408 / 1.431	-1.617	355 / 477	-0.228 (0.820)

The frozen short-horizon full-regression descriptive screen retains matched score gains across the three inspected windows and a changing rolling-reference direction. It is not confirmation after inspection or independent replication. The complete longer-horizon results also retain matched score gains, and the full regression's loss

against the rolling reference. A score gain can coexist with more breaches; neither a conditional-coverage certificate nor an economic operating benefit follows from these tables.

The added eight pairwise tests use only raw standard-normal two-sided DM-type working references, separate from all original adjusted families. For n paired losses at horizon H,

$$L = \min\Big(n-1, \max\Big\{H, \Big\lfloor 4(n/100)^{2/9} \Big\rfloor\Big\}\Big).$$

The centered Bartlett Newey–West covariance uses divisor n, and the statistic is the mean regression-minus-reference loss divided by $\sqrt{\widehat{\mathrm{LRV}}/n}$. Nonpositive long-run variance or constant loss differences retain an unavailable statistic and their specified status; there is no epsilon repair. These p-values receive no multiplicity or full selection adjustment and do not establish equivalence, real-time validity or independent confirmation. Missing old longer-horizon original-menu, parsimonious, rolling-DM or breach-count fields stay NA; the old five-menu rule is not relabelled as the original three-menu regression. No missing old test was recalculated.

S27 Earlier calibrated simulation and numerical failures

This distinct calibrated design uses constant-variance AR(1) levels, where the peak depends on the current position (D0), an hourly AR–GARCH–t law with a known hour shape (D1), and a weekly AR–GARCH–t law (D2). D0/D2 share the original pre-validation Treasury calibration; D1 uses the original pre-validation FX prefix. Levels are divided by the prefix standard deviation without demeaning. The frozen calibration, clocks, origin geometry, parameters and numerical rules are retained in the imported calibrated archive. The weekly clock, validation span, refit schedule and six-candidate menu, including the intercept-only limit, differ from Section S24. Their short-history outcomes cannot be treated as replications of one design.

Each completed cell reports repetition medians and quartiles of the within-repetition relative loss against F. These differ from the ratio of across-repetition mean losses in Section S24. Quartiles describe dispersion under fixed fitted laws, not confidence limits. The known-state oracle is a finite future-path approximation; its realized score need not beat F in every repetition. Ordinary-rank I, oracle results, strict breaches and intercept-only selections remain separate diagnostics.

Table S27: Legacy calibrated simulation: all 24 planned cells.

Process	History	Features	Complete / required	Regression Δ [IQR]	Win	Intercept change	Oracle change	Historical hit	Intercept hit	Regression hit	Oracle hit	Infinite penalty
D0	52	4	100/100	−0.2 [−4.6, 22.1]	0.510	+0.055	-30.0	0.046	0.064	0.129	0.048	0.110
D0	52	33	100/100	1.1 [−2.0, 6.1]	0.390	+0.055	-30.0	0.046	0.064	0.094	0.048	0.300
D0	260	4	100/100	−15.0 [−20.4, −7.3]	0.980	+0.030	-24.6	0.050	0.056	0.068	0.048	0.000
D0	260	33	100/100	−2.2 [−8.2, 2.7]	0.640	+0.030	-24.6	0.050	0.056	0.094	0.048	0.060
D0	720	4	100/100	−18.8 [−24.4, −14.5]	0.980	+0.046	-23.3	0.050	0.051	0.055	0.048	0.030
D0	720	33	100/100	−14.6 [−21.0, −9.7]	0.980	+0.046	-23.3	0.050	0.051	0.064	0.048	0.050
D0	2880	4	100/100	−21.8 [−25.2, −17.2]	0.990	-0.005	-23.6	0.050	0.050	0.048	0.048	0.020
D0	2880	33	100/100	−19.8 [−23.7, −15.2]	0.990	-0.005	-23.6	0.050	0.050	0.052	0.048	0.020
D1	52	4	10/100	**FAIL**	–	–	–	–	–	–	–	–
D1	52	33	100/100	0.2 [−1.1, 2.1]	0.470	+0.013	-34.0	0.083	0.096	0.096	0.055	0.770
D1	260	4	4/100	**FAIL**	–	–	–	–	–	–	–	–
D1	260	33	56/100	**FAIL**	–	–	–	–	–	–	–	–
D1	720	4	3/100	**FAIL**	–	–	–	–	–	–	–	–
D1	720	33	17/100	**FAIL**	–	–	–	–	–	–	–	–
D1	2880	4	3/100	**FAIL**	–	–	–	–	–	–	–	–
D1	2880	33	32/100	**FAIL**	–	–	–	–	–	–	–	–
D2	52	4	100/100	1.2 [−3.1, 19.8]	0.430	-0.485	-29.5	0.055	0.070	0.101	0.046	0.180
D2	52	33	100/100	0.4 [−1.6, 5.7]	0.460	-0.485	-29.5	0.055	0.070	0.098	0.046	0.270
D2	260	4	100/100	−13.9 [−20.5, −5.2]	0.910	-0.036	-25.6	0.052	0.056	0.069	0.046	0.030
D2	260	33	100/100	−0.8 [−7.8, 2.9]	0.600	-0.036	-25.6	0.052	0.056	0.097	0.046	0.080
D2	720	4	100/100	−18.6 [−22.6, −10.9]	0.980	-0.044	-24.8	0.046	0.048	0.054	0.046	0.030
D2	720	33	100/100	−12.4 [−18.8, −4.1]	0.910	-0.044	-24.8	0.046	0.048	0.059	0.046	0.050
D2	2880	4	100/100	−20.1 [−24.9, −15.3]	0.970	-0.017	-24.1	0.044	0.046	0.047	0.046	0.030
D2	2880	33	100/100	−20.6 [−24.1, −14.4]	0.950	-0.017	-24.1	0.044	0.046	0.052	0.046	0.080

Δ is the median relative loss change against F (percent); IQR is across registered repetitions. Win and hit columns are proportions; I is the ordinary empirical quantile and O is the finite nested oracle. The ∞ column is the frequency of selecting the intercept limit. All seven failed D1 rows remain unavailable after ADMM nonconvergence, even when some repetitions completed. D0: constant-variance AR(1) levels, where the peak depends on the current position. D1: FX-calibrated hourly; D2: weekly-calibrated. The broader legacy menu includes the intercept limit; these results are not the WP3 ratio-of-mean-losses.

Table S28: First recorded failure in each unavailable legacy cell.

Process	History	Features	Rep	Stage	λ	Saved reason
D1	2880	4	1	evaluation	0.001	QuantileFitError:NONCONVERGENCE
D1	720	4	3	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	260	4	4	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	52	4	10	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	720	33	17	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	2880	33	32	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	260	33	56	validation	0.001	QuantileFitError:NONCONVERGENCE

Repetition identifiers are zero based. Each event belongs to a whole failed cell in Table S27; no partial repetition summary is substituted. The original failure stage, penalty and reason are retained without a rerun.

A required numerical failure blocks the whole cell. Partial repetitions do not supply a replacement median, quartile, win rate or breach rate. The retained nonconvergence exceptions mean that the unchanged solver did not obtain its joint residual-and-gap acceptance certificate within its iteration budget. They establish neither nonexistence of an optimum nor poor predictive accuracy of an unavailable report. Final residuals and gaps were not retained; the four-feature exceptions also lack exact refit context. These quantities remain unknown. The original constant-variance and short-weekly predictions are unsupported; the hourly and history-order predictions are unassessable. These outcomes identify neither the financial loss mechanism nor a pure frequency effect.

S28 Pre-evaluation stability screen and switching stress simulation

The separately frozen stability-study protocol uses only pre-evaluation prefixes. Its joint OLS score includes intercept, AR slope and centered residual-square channels (Hansen, 1992). Each qualified whole-prefix AR–GARCH–t fit generates a plug-in parametric bootstrap reference. OLS and the score statistic are re-estimated on each bootstrap path; the generator is fitted once. Published iid critical values are not used. With the registered $B = 999$ bootstrap paths, the p value is $(1 + \#\{H_b \geq H_{\text{obs}}\})/(B + 1)$; the registered draw count and all exceedances remain in the archive. The table retains the observed statistic, bootstrap p value, Holm adjustment, moment qualification and descriptive cuts. Qualified fitted nulls require $\nu > 4$ and $q_4 = \alpha^2\{3(\nu - 2)/(\nu - 4)\} + 2\alpha\beta + \beta^2 < 1 - 10^{-8}$. A successful fit and this fourth-moment gate do not prove that the real law has those moments. The plug-in calibration supplies no exact finite-sample size guarantee; rejection can also reflect misspecification or outliers.

Table S29: Registered prefix stability diagnostics: all 15 source rows.

Source	Statistic	p	Holm	Moment valid	q_4	Cuts	Dates	Segments	h	Break cap	State
GS-B	1.83	0.051	0.306	Y	0.905	104/293	2015-04-08/2018-11-21	104/189/111	104	Y	OK
MBS-TOT	1.26	0.208	0.5	Y	0.948	189/293	2016-11-23/2018-11-21	189/104/111	104	Y	OK
CS-TOT	1.18	0.022	0.2	Y	0.0642	none	–	404	104	N	OK
FGS-TOT	4.07	0.001	0.013	Y	0.981	138	2015-12-02	138/266	104	N	OK
TIPS-L2	1.13	0.02	0.2	Y	0.983	107	2015-04-29	107/297	104	N	OK
TIPS-G11	0.901	0.063	0.315	Y	0.972	none	–	404	104	N	OK
TIPS-G2	1.28	0.018	0.198	Y	0.167	none	–	404	104	N	OK
TIPS-G6L11	2.51	0.001	0.013	Y	0.973	163	2016-05-25	163/241	104	N	OK
GSC-L2	1.15	0.213	0.5	Y	0.976	289	2018-10-24	289/115	104	N	OK
GSC-G2L3	2.09	0.155	0.5	Y	0.978	104/269	2015-04-08/2018-06-06	104/165/135	104	Y	OK
GSC-G3L6	2.78	0.024	0.2	Y	0.94	117/288	2015-07-08/2018-10-17	117/171/116	104	Y	OK
GSC-G6L7	0.799	0.125	0.5	Y	0.926	none	–	404	104	N	OK
GSC-G7L11	1.64	0.024	0.2	Y	0.937	145	2016-01-20	145/259	104	N	OK
FX daily	3.13	–	–	N	–	128/271	2014-07-06/2014-12-19	128/143/133	104	Y	NR
FX hourly	–	–	–	–	–	2036/5365	–	2036/3329/2743	1622	Y	OF

H: Hansen joint statistic; h: minimum segment transitions; M4: registered fitted-null fourth-moment qualification. Cuts use transition-row units; dates mark the first target after each cut. Unarchived dates remain – even when cuts are available. Source codes omit the common PDPOS prefix. NR: daily FX NONREGULAR_NULL (Student fourth moment unavailable); OF: hourly FX GARCH_OPTIMIZATION_FAILED. BF: a restricted-boundary fit; OK does not certify the model. Unavailable raw p and Holm values remain –; internal Holm placeholders are not tests. Segmentation is descriptive, and the two-break cap does not establish significant individual breaks.

Candidate cuts minimize a separate-variance Gaussian AR quasi-likelihood criterion with penalty $(3+4k)\log M$, where M counts transitions. The fixed search allows at most two cuts and requires each segment to contain at least $\max\{104, \lceil .20M \rceil\}$ transitions. These are descriptive calibration partitions, not individually significant breaks. Whole-prefix scaling is shared by every segment without demeaning; the first target in a segment keeps its preceding observed lag. The constrained AR and GARCH fits have one declared start and no retry. The daily nonregular null and hourly optimization failure retain unavailable primary p values; neither is evidence of stability.

The constant control retains the whole-prefix parameters. The stress arm cycles the fitted regimes with their original transition dwell lengths and a fixed repetition-phase grid. Level, innovation and variance state continue through each switch. The last-to-first wrap is artificial, rather than an observed historical break. Common uniforms are transformed through each arm's own standardized t law; a no-cut source reuses the identical whole-fit path. Both arms retain the forecasting procedures of Section S24. The following complete grids retain failed conditional cells and all F cells.

Table S30: Complete source-by-arm conditional comparison, $W = 52$.

Source	Arm	Full regression Δ [IQR]	Win	Hit	Four-feature Δ [IQR]	Win	Hit
GS-B	C	117.2 [44.3, 191.7]	0.094	0.312	7.2 [−6.6, 20.2]	0.344	0.097
GS-B	S	91.3 [20.0, 174.6]	0.188	0.278	3.2 [−6.6, 50.3]	0.406	0.090
MBS-TOT	C	79.8 [6.1, 214.8]	0.188	0.306	−0.9 [−8.3, 12.5]	0.500	0.097
MBS-TOT	S	140.8 [44.3, 333.9]	0.188	0.285	−0.4 [−6.7, 32.3]	0.531	0.069
CS-TOT	C	124.3 [38.5, 250.2]	0.125	0.340	−3.1 [−12.3, 16.2]	0.594	0.090
CS-TOT	S	124.3 [38.5, 250.2]	0.125	0.340	−3.1 [−12.3, 16.2]	0.594	0.090
FGS-TOT	C	201.0 [128.1, 302.5]	0.062	0.417	1.9 [−11.5, 16.2]	0.469	0.083
FGS-TOT	S	170.1 [27.7, 233.5]	0.062	0.354	−4.4 [−17.6, 12.7]	0.656	0.069
TIPS-L2	C	119.3 [37.1, 213.3]	0.031	0.285	−2.1 [−11.1, 7.6]	0.594	0.069
TIPS-L2	S	101.9 [45.6, 242.1]	0.062	0.319	−0.0 [−5.1, 6.8]	0.500	0.076
TIPS-G11	C	131.3 [62.5, 239.8]	0.125	0.333	1.1 [−4.2, 28.2]	0.406	0.111
TIPS-G11	S	131.3 [62.5, 239.8]	0.125	0.333	1.1 [−4.2, 28.2]	0.406	0.111
TIPS-G2	C	107.9 [47.3, 166.9]	0.125	0.306	−1.5 [−8.2, 9.6]	0.531	0.076
TIPS-G2	S	107.9 [47.3, 166.9]	0.125	0.306	−1.5 [−8.2, 9.6]	0.531	0.076
TIPS-G6L11	C	92.4 [56.4, 168.2]	0.031	0.312	0.1 [−4.3, 8.1]	0.500	0.104
TIPS-G6L11	S	121.4 [57.6, 217.0]	0.062	0.319	3.3 [−3.6, 13.9]	0.406	0.097
GSC-L2	C	116.2 [16.6, 211.1]	0.156	0.285	−0.0 [−5.7, 12.2]	0.500	0.090
GSC-L2	S	123.1 [54.3, 221.6]	0.062	0.306	−2.5 [−7.3, 4.8]	0.656	0.069
GSC-G2L3	C	112.9 [53.6, 164.0]	0.031	0.271	−0.4 [−3.4, 3.3]	0.531	0.090
GSC-G2L3	S	109.1 [52.7, 218.1]	0.062	0.299	10.6 [−8.7, 44.5]	0.438	0.097
GSC-G3L6	C	135.1 [36.7, 237.1]	0.156	0.306	2.8 [−1.3, 14.7]	0.375	0.097
GSC-G3L6	S	166.5 [122.1, 242.0]	0.031	0.368	5.3 [−1.0, 66.4]	0.375	0.104
GSC-G6L7	C	90.8 [30.2, 191.6]	0.156	0.278	1.7 [−3.4, 10.4]	0.438	0.097
GSC-G6L7	S	90.8 [30.2, 191.6]	0.156	0.278	1.7 [−3.4, 10.4]	0.438	0.097
GSC-G7L11	C	92.9 [56.3, 208.5]	0.125	0.243	−4.7 [−9.8, 3.6]	0.688	0.062
GSC-G7L11	S	80.1 [40.1, 196.3]	0.031	0.264	−2.0 [−10.0, 11.9]	0.531	0.083

C: fitted constant control; S: parameter-switching stress. Source codes omit PDPOS. Δ: median percent relative loss against the matched F across all 32 registered repetitions, with IQR. Win and QR hit are proportions. FAIL 31/32 is a whole failed cell, with no partial median or win/hit summary; all mandatory repeats must succeed. The failed QR33 W260 cells are PDPOSTIPS-G11, repetition 24, in both arms, caused by validation NONCONVERGENCE. QR4 may select a completed candidate under the unchanged registered policy; its failed candidates remain in the archive.

Table S31: Complete source-by-arm conditional comparison, $W = 260$.

Source	Arm	Full regression Δ [IQR]	Win	Hit	Four-feature Δ [IQR]	Win	Hit
GS-B	C	2.9 [−11.6, 23.1]	0.438	0.069	−2.0 [−16.4, 2.1]	0.594	0.042
GS-B	S	4.6 [−12.9, 57.1]	0.438	0.104	0.2 [−3.6, 5.0]	0.500	0.062
MBS-TOT	C	7.0 [−9.0, 25.6]	0.406	0.076	−5.0 [−15.3, 0.4]	0.688	0.021
MBS-TOT	S	18.8 [2.5, 50.1]	0.250	0.062	−0.1 [−2.6, 15.8]	0.500	0.035
CS-TOT	C	2.7 [−12.5, 39.2]	0.500	0.125	−1.2 [−9.9, 17.8]	0.594	0.076
CS-TOT	S	2.7 [−12.5, 39.2]	0.500	0.125	−1.2 [−9.9, 17.8]	0.594	0.076
FGS-TOT	C	13.8 [−7.5, 45.0]	0.344	0.104	−3.0 [−14.1, 7.0]	0.688	0.062

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Source	Arm	Full regression Δ [IQR]	Win	Hit	Four-feature Δ [IQR]	Win	Hit
FGS-TOT	S	15.5 [−17.3, 80.9]	0.438	0.076	0.2 [−3.6, 6.5]	0.438	0.062
TIPS-L2	C	9.3 [−14.2, 24.2]	0.469	0.104	−1.7 [−7.9, 10.6]	0.594	0.056
TIPS-L2	S	8.9 [−5.9, 29.8]	0.344	0.069	−0.3 [−2.0, 2.1]	0.594	0.049
TIPS-G11	C	**FAIL** 31/32	–	–	−0.5 [−9.1, 10.7]	0.531	0.090
TIPS-G11	S	**FAIL** 31/32	–	–	−0.5 [−9.1, 10.7]	0.531	0.090
TIPS-G2	C	13.1 [−11.1, 46.3]	0.406	0.111	−0.3 [−8.9, 4.1]	0.562	0.062
TIPS-G2	S	13.1 [−11.1, 46.3]	0.406	0.111	−0.3 [−8.9, 4.1]	0.562	0.062
TIPS-G6L11	C	15.6 [−3.7, 55.9]	0.344	0.125	−0.5 [−8.8, 13.9]	0.531	0.069
TIPS-G6L11	S	20.9 [−1.9, 62.9]	0.281	0.132	1.2 [−0.4, 10.0]	0.375	0.056
GSC-L2	C	10.4 [−14.0, 42.4]	0.344	0.069	−0.8 [−5.8, 2.7]	0.625	0.035
GSC-L2	S	24.5 [−2.6, 51.8]	0.312	0.083	0.8 [−0.9, 11.2]	0.406	0.021
GSC-G2L3	C	1.4 [−12.9, 13.2]	0.469	0.118	−1.4 [−16.3, 0.5]	0.688	0.083
GSC-G2L3	S	−0.6 [−17.0, 23.3]	0.500	0.104	−3.8 [−9.3, 3.4]	0.625	0.083
GSC-G3L6	C	−1.4 [−17.6, 24.4]	0.562	0.104	−1.3 [−6.3, 0.7]	0.688	0.056
GSC-G3L6	S	13.3 [−7.7, 30.5]	0.312	0.097	0.3 [−0.8, 4.6]	0.438	0.049
GSC-G6L7	C	9.1 [−7.4, 39.0]	0.438	0.097	−0.7 [−3.5, 4.1]	0.594	0.056
GSC-G6L7	S	9.1 [−7.4, 39.0]	0.438	0.097	−0.7 [−3.5, 4.1]	0.594	0.056
GSC-G7L11	C	3.6 [−10.7, 30.2]	0.438	0.076	−0.5 [−5.5, 1.9]	0.531	0.042
GSC-G7L11	S	−3.4 [−19.0, 11.7]	0.562	0.056	−5.2 [−20.8, 3.2]	0.625	0.062

C: fitted constant control; S: parameter-switching stress. Source codes omit PDPOS. Δ: median percent relative loss against the matched F across all 32 registered repetitions, with IQR. Win and QR hit are proportions. FAIL 31/32 is a whole failed cell, with no partial median or win/hit summary; all mandatory repeats must succeed. The failed QR33 W260 cells are PDPOSTIPS-G11, repetition 24, in both arms, caused by validation NONCONVERGENCE. QR4 may select a completed candidate under the unchanged registered policy; its failed candidates remain in the archive.

Table S32: Corrected-rank F benchmark: all 52 source-by-arm-history cells.

Source	Arm	History	Complete / required	Loss [IQR]	Hit [IQR]
GS-B	C	52	32/32	0.09015 [0.07863, 0.1076]	0.05556 [0.02778, 0.09722]
GS-B	C	260	32/32	0.07989 [0.06798, 0.09176]	0.04167 [0, 0.09722]
GS-B	S	52	32/32	0.08146 [0.06128, 0.1055]	0.05556 [0.02778, 0.08333]
GS-B	S	260	32/32	0.08697 [0.0705, 0.1085]	0.04167 [0, 0.1528]
MBS-TOT	C	52	32/32	0.05759 [0.0481, 0.07073]	0.05556 [0, 0.08333]
MBS-TOT	C	260	32/32	0.05332 [0.04593, 0.05875]	0.02083 [0, 0.05903]
MBS-TOT	S	52	32/32	0.0464 [0.04173, 0.06648]	0.03472 [0, 0.08333]
MBS-TOT	S	260	32/32	0.04574 [0.04135, 0.05452]	0.01389 [0, 0.05556]
CS-TOT	C	52	32/32	0.08467 [0.07035, 0.1165]	0.04167 [0, 0.09722]
CS-TOT	C	260	32/32	0.08322 [0.06532, 0.1016]	0.04861 [0, 0.08681]
CS-TOT	S	52	32/32	0.08467 [0.07035, 0.1165]	0.04167 [0, 0.09722]
CS-TOT	S	260	32/32	0.08322 [0.06532, 0.1016]	0.04861 [0, 0.08681]
FGS-TOT	C	52	32/32	0.05269 [0.04771, 0.06046]	0.04861 [0.01389, 0.06944]
FGS-TOT	C	260	32/32	0.04924 [0.0459, 0.05809]	0.04167 [0.01389, 0.06944]
FGS-TOT	S	52	32/32	0.06601 [0.05181, 0.07969]	0.02778 [0.01389, 0.07292]
FGS-TOT	S	260	32/32	0.05653 [0.05491, 0.07435]	0.04167 [0, 0.1111]
TIPS-L2	C	52	32/32	0.09575 [0.07975, 0.1155]	0.04167 [0.02778, 0.07292]
TIPS-L2	C	260	32/32	0.0859 [0.07622, 0.09682]	0.04167 [0.02431, 0.08333]
TIPS-L2	S	52	32/32	0.08608 [0.0697, 0.111]	0.05556 [0.02778, 0.08333]
TIPS-L2	S	260	32/32	0.07837 [0.06901, 0.1028]	0.04167 [0.01389, 0.08681]
TIPS-G11	C	52	32/32	0.09022 [0.06905, 0.1083]	0.05556 [0.01389, 0.08333]
TIPS-G11	C	260	32/32	0.08016 [0.07022, 0.1084]	0.05556 [0.02778, 0.1111]
TIPS-G11	S	52	32/32	0.09022 [0.06905, 0.1083]	0.05556 [0.01389, 0.08333]
TIPS-G11	S	260	32/32	0.08016 [0.07022, 0.1084]	0.05556 [0.02778, 0.1111]
TIPS-G2	C	52	32/32	0.1007 [0.07528, 0.121]	0.05556 [0.01389, 0.08681]
TIPS-G2	C	260	32/32	0.08534 [0.06626, 0.1093]	0.05556 [0.02431, 0.08681]
TIPS-G2	S	52	32/32	0.1007 [0.07528, 0.121]	0.05556 [0.01389, 0.08681]
TIPS-G2	S	260	32/32	0.08534 [0.06626, 0.1093]	0.05556 [0.02431, 0.08681]
TIPS-G6L11	C	52	32/32	0.09185 [0.08051, 0.1171]	0.07639 [0.05208, 0.09722]
TIPS-G6L11	C	260	32/32	0.0826 [0.07391, 0.09724]	0.0625 [0.02778, 0.1007]
TIPS-G6L11	S	52	32/32	0.09133 [0.07505, 0.1247]	0.06944 [0.03819, 0.09722]
TIPS-G6L11	S	260	32/32	0.08318 [0.07664, 0.1058]	0.04167 [0, 0.08333]
GSC-L2	C	52	32/32	0.04554 [0.04079, 0.05505]	0.04861 [0.02431, 0.08681]
GSC-L2	C	260	32/32	0.04131 [0.0388, 0.04593]	0.02778 [0, 0.08333]
GSC-L2	S	52	32/32	0.04493 [0.0366, 0.04898]	0.04861 [0.01389, 0.08333]
GSC-L2	S	260	32/32	0.03859 [0.03435, 0.04521]	0.02778 [0, 0.06944]

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Source	Arm	History	Complete / required	Loss [IQR]	Hit [IQR]
GSC-G2L3	C	52	32/32	0.06574 [0.05149, 0.0898]	0.05556 [0.02778, 0.08333]
GSC-G2L3	C	260	32/32	0.0645 [0.05369, 0.08905]	0.06944 [0.01389, 0.1111]
GSC-G2L3	S	52	32/32	0.06149 [0.05254, 0.08228]	0.04861 [0.01389, 0.08681]
GSC-G2L3	S	260	32/32	0.0616 [0.05365, 0.0779]	0.04861 [0, 0.08333]
GSC-G3L6	C	52	32/32	0.05913 [0.05199, 0.08991]	0.05556 [0.01389, 0.1111]
GSC-G3L6	C	260	32/32	0.05448 [0.04917, 0.06724]	0.04167 [0.02431, 0.07292]
GSC-G3L6	S	52	32/32	0.05589 [0.04396, 0.07093]	0.07639 [0.02778, 0.1111]
GSC-G3L6	S	260	32/32	0.04793 [0.045, 0.05747]	0.04861 [0, 0.09722]
GSC-G6L7	C	52	32/32	0.05322 [0.04505, 0.06819]	0.04167 [0.01389, 0.07292]
GSC-G6L7	C	260	32/32	0.0482 [0.04246, 0.05957]	0.04861 [0.02431, 0.06944]
GSC-G6L7	S	52	32/32	0.05322 [0.04505, 0.06819]	0.04167 [0.01389, 0.07292]
GSC-G6L7	S	260	32/32	0.0482 [0.04246, 0.05957]	0.04861 [0.02431, 0.06944]
GSC-G7L11	C	52	32/32	0.05278 [0.04627, 0.06312]	0.02778 [0.01389, 0.06944]
GSC-G7L11	C	260	32/32	0.04761 [0.04315, 0.05204]	0.02778 [0, 0.05903]
GSC-G7L11	S	52	32/32	0.05369 [0.04537, 0.06031]	0.04167 [0, 0.08333]
GSC-G7L11	S	260	32/32	0.04759 [0.04422, 0.05826]	0.04861 [0.01042, 0.1007]

C: fitted constant control; S: parameter-switching stress. Source codes omit PDPOS. Loss is the median across repetitions of the within-repetition mean pinball score, with IQR; Hit is the corresponding breach-rate median and IQR. All 52 cells have 32/32 valid final F receipts. Loss units follow each source-specific whole-prefix SD normalization; different sources are not pooled. No failed-cell partial median is substituted.

Loss summaries are medians across repetitions within each source, then equal- weight medians across every required source. They are not the controlled grid's ratio-of-means estimand. A required repetition failure makes its whole cell unavailable; successful subsets do not replace missing full-grid results. The raw weekly screening threshold is met, while multiplicity adjustment retains fewer rejections. The near-parity constant-control prediction is unsupported. The two-history switching prediction and the daily/hourly comparison are unassessable. Constant controls themselves generate large losses, so instability is not necessary for losses in these procedures. The empirical low-frequency mechanism remains unidentified.

S29 Paired menu isolation: full source inventory

Both QR variants use penalties $\{.001, .01, .1, 1, 10, \infty\}$; the infinite candidate is ordinary intercept-only I, distinct from corrected-rank F. Full regression requires every candidate to complete. Four-predictor regression selects from its completed eligible candidates, as before. Exact ties now prefer stronger shrinkage in both variants; no actual exact tie occurred. Old and expanded arms use identical source calibration, random streams, path geometry and forecasting clocks. The switching-phase denominator remains thirty-two. Two pilot repetitions selected the full thirty-two formal repetitions before launch. The simulated feature clock is inherited hourly geometry despite weekly calibration parameters; it is not a literal weekly-calendar replay.

The design preceded every new path calculation. M1 requires the constant-arm history-52 full-regression median within five percent of F. M2 requires the switching median at least five percentage points higher. M3 requires both aggregate improvements and at least twelve jointly improving sources. M1 and M3 hold; M2 fails. Full history-260 regression summaries remain unavailable after the required TIPS-G11 repetition fails in both arms. Successful repetitions or twelve-source subsets never replace them.

Table S33: Every menu-isolation source/arm/history/method cell. Relative-loss medians use all required repetitions.

Source	Arm	History	Predictors	Original median	Expanded median	Complete/required	State
GS-B	C	52	33	117.16	4.24	32/32	COMPLETE
GS-B	C	52	4	7.20	8.70	32/32	COMPLETE
GS-B	C	260	33	2.90	0.10	32/32	COMPLETE
GS-B	C	260	4	-2.02	-1.92	32/32	COMPLETE
GS-B	S	52	33	91.35	2.47	32/32	COMPLETE
GS-B	S	52	4	3.22	3.17	32/32	COMPLETE
GS-B	S	260	33	4.56	0.56	32/32	COMPLETE
GS-B	S	260	4	0.23	0.09	32/32	COMPLETE
MBS-TOT	C	52	33	79.84	-0.49	32/32	COMPLETE
MBS-TOT	C	52	4	-0.87	-0.70	32/32	COMPLETE
MBS-TOT	C	260	33	7.02	0.97	32/32	COMPLETE
MBS-TOT	C	260	4	-5.00	-5.15	32/32	COMPLETE
MBS-TOT	S	52	33	140.83	-0.59	32/32	COMPLETE
MBS-TOT	S	52	4	-0.44	-0.92	32/32	COMPLETE
MBS-TOT	S	260	33	18.77	0.91	32/32	COMPLETE
MBS-TOT	S	260	4	-0.06	-0.22	32/32	COMPLETE
CS-TOT	C	52	33	124.34	-3.01	32/32	COMPLETE
CS-TOT	C	52	4	-3.07	-3.07	32/32	COMPLETE
CS-TOT	C	260	33	2.68	-0.19	32/32	COMPLETE
CS-TOT	C	260	4	-1.21	-1.13	32/32	COMPLETE
CS-TOT	S	52	33	124.34	-3.01	32/32	COMPLETE
CS-TOT	S	52	4	-3.07	-3.07	32/32	COMPLETE

Source	Arm	History	Predictors	Original median	Expanded median	Complete/required	State
CS-TOT	S	260	33	2.68	-0.19	32/32	COMPLETE
CS-TOT	S	260	4	-1.21	-1.13	32/32	COMPLETE
FGS-TOT	C	52	33	201.01	4.44	32/32	COMPLETE
FGS-TOT	C	52	4	1.90	1.90	32/32	COMPLETE
FGS-TOT	C	260	33	13.76	1.34	32/32	COMPLETE
FGS-TOT	C	260	4	-2.98	-2.98	32/32	COMPLETE
FGS-TOT	S	52	33	170.12	-0.23	32/32	COMPLETE
FGS-TOT	S	52	4	-4.36	-4.36	32/32	COMPLETE
FGS-TOT	S	260	33	15.48	-0.40	32/32	COMPLETE
FGS-TOT	S	260	4	0.23	0.48	32/32	COMPLETE
TIPS-L2	C	52	33	119.29	-0.54	32/32	COMPLETE
TIPS-L2	C	52	4	-2.10	-2.10	32/32	COMPLETE
TIPS-L2	C	260	33	9.32	0.64	32/32	COMPLETE
TIPS-L2	C	260	4	-1.69	-1.69	32/32	COMPLETE
TIPS-L2	S	52	33	101.92	0.53	32/32	COMPLETE
TIPS-L2	S	52	4	-0.03	0.38	32/32	COMPLETE
TIPS-L2	S	260	33	8.88	-0.38	32/32	COMPLETE
TIPS-L2	S	260	4	-0.27	-0.28	32/32	COMPLETE
TIPS-G11	C	52	33	131.26	0.98	32/32	COMPLETE
TIPS-G11	C	52	4	1.11	1.14	32/32	COMPLETE
TIPS-G11	C	260	33	NA	NA	31/32	FAILED
TIPS-G11	C	260	4	-0.48	-0.49	32/32	COMPLETE
TIPS-G11	S	52	33	131.26	0.98	32/32	COMPLETE
TIPS-G11	S	52	4	1.11	1.14	32/32	COMPLETE
TIPS-G11	S	260	33	NA	NA	31/32	FAILED
TIPS-G11	S	260	4	-0.48	-0.49	32/32	COMPLETE
TIPS-G2	C	52	33	107.85	-4.00	32/32	COMPLETE
TIPS-G2	C	52	4	-1.55	-1.69	32/32	COMPLETE
TIPS-G2	C	260	33	13.09	0.12	32/32	COMPLETE
TIPS-G2	C	260	4	-0.29	-0.43	32/32	COMPLETE
TIPS-G2	S	52	33	107.85	-4.00	32/32	COMPLETE
TIPS-G2	S	52	4	-1.55	-1.69	32/32	COMPLETE
TIPS-G2	S	260	33	13.09	0.12	32/32	COMPLETE
TIPS-G2	S	260	4	-0.29	-0.43	32/32	COMPLETE
TIPS-G6L11	C	52	33	92.36	1.00	32/32	COMPLETE
TIPS-G6L11	C	52	4	0.11	0.44	32/32	COMPLETE
TIPS-G6L11	C	260	33	15.63	2.53	32/32	COMPLETE
TIPS-G6L11	C	260	4	-0.54	-0.64	32/32	COMPLETE
TIPS-G6L11	S	52	33	121.40	3.18	32/32	COMPLETE
TIPS-G6L11	S	52	4	3.27	3.68	32/32	COMPLETE
TIPS-G6L11	S	260	33	20.88	1.90	32/32	COMPLETE
TIPS-G6L11	S	260	4	1.23	1.23	32/32	COMPLETE
GSC-L2	C	52	33	116.24	0.21	32/32	COMPLETE
GSC-L2	C	52	4	-0.02	1.85	32/32	COMPLETE
GSC-L2	C	260	33	10.40	-0.21	32/32	COMPLETE
GSC-L2	C	260	4	-0.80	-1.22	32/32	COMPLETE
GSC-L2	S	52	33	123.05	-1.88	32/32	COMPLETE
GSC-L2	S	52	4	-2.47	-2.17	32/32	COMPLETE
GSC-L2	S	260	33	24.51	0.08	32/32	COMPLETE
GSC-L2	S	260	4	0.81	0.87	32/32	COMPLETE
GSC-G2L3	C	52	33	112.85	0.74	32/32	COMPLETE
GSC-G2L3	C	52	4	-0.43	-0.28	32/32	COMPLETE
GSC-G2L3	C	260	33	1.42	0.36	32/32	COMPLETE
GSC-G2L3	C	260	4	-1.39	-1.02	32/32	COMPLETE
GSC-G2L3	S	52	33	109.05	0.02	32/32	COMPLETE
GSC-G2L3	S	52	4	10.61	10.73	32/32	COMPLETE
GSC-G2L3	S	260	33	-0.63	-1.22	32/32	COMPLETE
GSC-G2L3	S	260	4	-3.84	-3.95	32/32	COMPLETE
GSC-G3L6	C	52	33	135.13	0.90	32/32	COMPLETE
GSC-G3L6	C	52	4	2.78	2.43	32/32	COMPLETE
GSC-G3L6	C	260	33	-1.44	-1.16	32/32	COMPLETE
GSC-G3L6	C	260	4	-1.26	-1.37	32/32	COMPLETE
GSC-G3L6	S	52	33	166.48	4.12	32/32	COMPLETE
GSC-G3L6	S	52	4	5.26	6.42	32/32	COMPLETE
GSC-G3L6	S	260	33	13.34	1.34	32/32	COMPLETE
GSC-G3L6	S	260	4	0.31	0.61	32/32	COMPLETE
GSC-G6L7	C	52	33	90.84	-0.32	32/32	COMPLETE
GSC-G6L7	C	52	4	1.73	2.14	32/32	COMPLETE
GSC-G6L7	C	260	33	9.12	0.02	32/32	COMPLETE
GSC-G6L7	C	260	4	-0.66	-0.75	32/32	COMPLETE
GSC-G6L7	S	52	33	90.84	-0.32	32/32	COMPLETE
GSC-G6L7	S	52	4	1.73	2.14	32/32	COMPLETE
GSC-G6L7	S	260	33	9.12	0.02	32/32	COMPLETE
GSC-G6L7	S	260	4	-0.66	-0.75	32/32	COMPLETE
GSC-G7L11	C	52	33	92.90	-0.52	32/32	COMPLETE
GSC-G7L11	C	52	4	-4.71	-4.35	32/32	COMPLETE
GSC-G7L11	C	260	33	3.61	-0.36	32/32	COMPLETE
GSC-G7L11	C	260	4	-0.51	-0.55	32/32	COMPLETE
GSC-G7L11	S	52	33	80.11	-2.78	32/32	COMPLETE
GSC-G7L11	S	52	4	-2.03	-2.67	32/32	COMPLETE
GSC-G7L11	S	260	33	-3.41	0.20	32/32	COMPLETE
GSC-G7L11	S	260	4	-5.15	-5.15	32/32	COMPLETE

Constant=C; switching=S. Positive percentages mean higher loss than F. NA retains the whole failed cell; source medians and old results remain separate.

S30 CFTC sources, eligibility and information dates

The source is the CFTC TFF futures-only historical backcast for 2006–2016 plus annual 2017–2026 files. The retained observations begin in 2010. All eleven original source archives remain public; no other new financial data were acquired.

A complete generated calendar contains 877 Monday-labelled report weeks, from 2010-01-04 through 2026-10-19. Economic position dates are retained separately. A code needs one valid observation per week, a stable exchange and stable contract units. Missing weeks, duplicates and changes disqualify rather than interpolate or splice a contract. Known consolidation prefixes 13874,20974,12460 are excluded before ranking. This conservative rule is broader than the documented plus-suffixed codes. Full-period mean open interest ranks survivors, with literal-code ties. Seventeen of 141 observed codes qualify; the top-twenty cap never changes that count. The complete code-level eligibility and exclusions preserve every screening decision.

Validation indices are 553–604; evaluation indices are 611–870. All labels use horizon six, leaving six unissued origins between the segments. Validation position dates are 2020-08-11–2021-08-03; its last label ends 2021-09-14. Evaluation position dates are 2021-09-21–2026-09-08; its last label ends 2026-10-20. Training labels at refit r span r-H-W+1 through r-H. Monday labels index the inherited feature calendar; they do not claim Monday public availability. Actual positions are normally measured Tuesday and released Friday at 15:30 Eastern. Holiday, shutdown and ION-related publication delays prevent a first-release interpretation.

The selected category series are listed below. Net positions equal dealer long minus short in raw contract counts. Equal-weight median percentage changes preserve within-contract units; they are not common-dollar portfolio risk.

Code	Latest contract name	Mean open interest	Weeks
044601	UST 5Y NOTE - CHICAGO BOARD OF TRADE	3479174.2	877
043602	UST 10Y NOTE - CHICAGO BOARD OF TRADE	3344673.4	877
042601	UST 2Y NOTE - CHICAGO BOARD OF TRADE	2170647.9	877
045601	FED FUNDS - CHICAGO BOARD OF TRADE	1282757.7	877
020601	UST BOND - CHICAGO BOARD OF TRADE	999944.0	877
099741	EURO FX - CHICAGO MERCANTILE EXCHANGE	503616.5	877
1170E1	VIX FUTURES - CBOE FUTURES EXCHANGE	356897.7	877
097741	JAPANESE YEN - CHICAGO MERCANTILE EXCHANGE	210863.1	877
096742	BRITISH POUND - CHICAGO MERCANTILE EXCHANGE	197853.5	877
095741	MEXICAN PESO - CHICAGO MERCANTILE EXCHANGE	174995.0	877
090741	CANADIAN DOLLAR - CHICAGO MERCANTILE EXCHANGE	166229.3	877
232741	AUSTRALIAN DOLLAR - CHICAGO MERCANTILE EXCHANGE	154802.4	877
33874A	E-MINI S&P 400 STOCK INDEX - CHICAGO MERCANTILE EXCHANGE	75156.1	877
240743	NIKKEI STOCK AVERAGE YEN DENOM - CHICAGO MERCANTILE EXCHANGE	59766.6	877
092741	SWISS FRANC - CHICAGO MERCANTILE EXCHANGE	57197.3	877
098662	USD INDEX - ICE FUTURES U.S.	52473.9	877
112741	NZ DOLLAR - CHICAGO MERCANTILE EXCHANGE	44292.0	877

The six methods are corrected-rank F, ordinary I, four- and thirty-three-predictor regression, and finite delayed ACI-style F/QR33. Both regressions use the six-point menu. The paired ACI wrappers retain the existing finite calibration with delayed matured feedback; no asymptotic coverage claim is attached. The complete 204 method cells finish without a numerical retry. The full cell CSV and trace ZIP retain absolute scores, breaches, selected penalties and gamma values.

For the CFTC delayed wrappers, gamma belongs to $\{.001, .005, .01, .05\}$. Completed candidates rank by validation coverage-distance, then pinball loss, then smaller gamma. Initialization requires at least nineteen matured scores. Updates wait for the full six-record label, and evaluation carries the validation state. The six unissued gap origins create no artificial reports or feedback events. These settings accompany histories 52/260, 52 validation origins, 260 evaluation origins and 13-record refits; none is retuned after evaluation.

Two narrow official-format clarifications follow the original design and acquisition but precede every forecast. The first accepts only the official midnight suffix after the date; the second accepts integer counts written with an all-zero decimal suffix. Neither changes a mathematical count, sample threshold, screen or forecast setting. Original failed technical preparation receipts remain visible. This is not a claim of fully observation-blind ingestion design.

S31 Complete model confidence sets

The registered ninety-percent MCS uses the range statistic and 1,000 stationary-bootstrap draws, with block length $\max\{H, \lceil n^{1/3} \rceil\}$ in arch 8.0.0. Deterministic source/grid seeds and method order are recorded in the design. Every claimed complete report uses its entire declared origin grid and common target. Missing stored losses are derived only from saved reports, without issuing or fitting forecasts. Every one of 98 comparisons completes. The 60 known unavailable slots remain unavailable, not statistical exclusions.

Distinct procedure labels retain their classification even when a selected penalty is infinite or reports coincide. Exact loss duplicates enter the library once, then recover the same p-value and inclusion partition for all members. Conditional slots use their declared procedure identity. Matched F/I membership, literal uniqueness and unique-loss-class membership differ. Across the entire inventory, 90/98 sets contain matched F/I, 2/98 have a sole literal F/I member, and 4/98 have a sole loss class containing F/I. Conditional exclusion is 197/683 complete slots. Related sources and overlapping tasks provide no independent cross-grid test or simultaneous future guarantee.

Complete registered MCS membership and p-values. All 98 source/period/H/W comparisons are retained. Bold p-values denote included procedures using the saved literal rule $p > 0.10$; ordinary type denotes exclusion. Values are rounded to three decimals only for display. Membership uses unrounded saved values; exact p-values are public in `results/v10/mcs/report-values.json`. NA denotes a known unavailable procedure, never a statistical exclusion; — means that procedure was not registered in that particular grid. Multi-member exact-loss classes carry a superscript equal to the representative method ID on *every* member, including the representative. Unmarked values are singleton classes. Method IDs are display aliases only; all full labels and classifications appear in the final legend. The registered procedure uses 1000 stationary-bootstrap replications, confidence 0.90, statistic R, and block length $\max\{H, \lceil n^{1/3} \rceil\}$. Analytic identical-loss or single-method cases retain their saved status; no new inference is performed for this table.

FX original. Each method column is shown once for every row of this group.

Table S35: FX original; method block 1/3. Full-grid $n = 6625$.

Source	Horizon	History	Hist. headroom	Intercept only	Cal. SAV history	Cal. SAV regression	SAV headroom	Expanded full reg.	Endpoint plug-in	Four-feature Legacy multi-fold	Finite ridge	Expanded ridge	One-SE ridge
fx_hourly	6	720	0.000	0.002^{M02}	0.000	0.019	0.000	0.040^{M10}	0.000	0.010	NA	0.040^{M10}	0.040^{M10}	0.002^{M02}
fx_hourly	6	2880	0.000	0.000^{M02}	0.000	**0.385**	0.000	**1.000**^{M10}	0.000	0.001	—	**1.000**^{M10}	**1.000**^{M10}	0.000^{M02}
fx_hourly	24	720	**0.696**	**0.760**	0.009	0.000	**0.792**	0.009^{M10}	0.011	**0.250**	NA	0.009^{M10}	0.009^{M10}	0.009^{M10}
fx_hourly	24	2880	**0.360**	**0.360**	**0.360**	**0.360**	**0.360**	**0.360**^{M10}	0.005	**0.360**	—	**0.360**^{M10}	**0.360**^{M10}	0.078^{M16}

Table S36: FX original; method block 2/3. Full-grid $n = 6625$.

Source	Horizon	History	Multi-fold one-SE	Filtered paths	Peak-tail fit	Quantile forest	Nearest states	Original full reg.	Scaled neighbors	Frozen peak	Frozen transl.	State headroom	State peak	State transl.
fx_hourly	6	720	—	NA	NA	0.002	0.014	0.040^{M10}	0.016	0.000	0.004	0.002	0.000	0.014
fx_hourly	6	2880	**1.000**^{M10}	0.000	0.000	0.001	0.054	**1.000**^{M10}	0.039	0.000	0.000	—	—	—
fx_hourly	24	720	—	NA	NA	**0.199**	**0.445**	0.009^{M10}	**0.347**	0.000	**0.696**	**0.792**	0.067	**0.792**
fx_hourly	24	2880	0.078^{M16}	0.079	0.002	**0.153**	0.078	**0.360**^{M10}	0.052	0.000	**0.360**	—	—	—

Table S37: FX original; method block 3/3. Full-grid $n = 6625$.

Source	Horizon	History	Rolling headroom	Rolling peak	Rolling transl.	Mean combination	Median combination
fx_hourly	6	720	0.002	0.000	0.003	**1.000**	0.047
fx_hourly	6	2880	0.000	0.000	0.000	0.039	0.057
fx_hourly	24	720	**0.792**	0.000	**1.000**	**0.696**	**0.551**
fx_hourly	24	2880	**0.360**	0.000	**0.360**	**1.000**	**0.360**

FX early2015. Each method column is shown once for every row of this group.

Table S38: FX early2015; method block 1/1. Full-grid $n = 1440$.

Source	Horizon	History	Hist. headroom	Intercept only	Legacy multi-fold	Finite ridge	Expanded ridge	One-SE ridge	Multi-fold one-SE	Original full reg.	Original roll. hist.
fx_hourly	6	720	0.063	**0.675**	NA	**0.675**^{M14}	**0.675**^{M14}	**0.628**	—	—	**1.000**
fx_hourly	6	2880	0.000	0.000	—	**1.000**^{M14}	**1.000**^{M14}	0.009	**1.000**^{M14}	**1.000**^{M14}	—
fx_hourly	24	720	0.014	**0.242**	NA	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	—	—	—
fx_hourly	24	2880	0.000	**0.910**	—	**1.000**^{M14}	**1.000**^{M14}	**0.991**^{M16}	**0.991**^{M16}	—	—

Treasury bills. Each method column is shown once for every row of this group.

Table S39: Treasury bills; method block 1/3. Full-grid $n = 277$.

Source	Horizon	History	Hist. headroom	Intercept only	Cal. SAV history	Cal. SAV regression	SAV headroom	Expanded full reg.	Endpoint plug-in	Four-feature	Finite ridge	Expanded ridge	One-SE ridge	Multi-fold one-SE
nyfed_weekly	6	52	**0.809**^{M01}	**0.400**^{M02}	0.000	0.004	0.045	**0.940**^{M10}	**0.810**	**0.770**	**0.940**^{M10}	**0.400**^{M02}	**0.400**^{M02}	**0.400**^{M02}
nyfed_weekly	6	260	**0.663**	0.073	**0.144**	**0.194**	**0.273**	**0.452**^{M10}	**0.982**	**0.622**	**0.452**^{M10}	**0.452**^{M10}	**0.622**^{M16}	**0.622**^{M16}
nyfed_weekly	24	52	**0.862**^{M01}	**0.626**^{M02}	**0.926**	0.004	**0.253**	**0.925**^{M10}	**0.211**	**0.150**	**0.925**^{M10}	**0.626**^{M02}	**0.626**^{M02}	**0.626**^{M02}
nyfed_weekly	24	260	0.066	**0.165**^{M02}	NA	NA	**0.261**	**0.186**^{M10}	**0.186**	**0.178**	**0.186**^{M10}	**0.186**^{M10}	**0.165**^{M02}	**0.165**^{M02}

Table S40: Treasury bills; method block 2/3. Full-grid $n = 277$.

Source	Horizon	History	Filtered paths	Peak-tail fit	Quantile forest	Nearest states	Original full reg.	Scaled neighbors	Frozen peak	Frozen transl.	State headroom	State peak	State transl.	Rolling headroom
nyfed_weekly	6	52	**0.940**	NA	**0.940**	**0.809**^{M01}	0.012	0.016	**0.743**	**0.767**	**0.863**	**0.940**	**0.802**	**0.597**
nyfed_weekly	6	260	**0.897**	**0.982**	**0.622**	**0.845**	0.028	**0.157**	**0.982**	**0.687**	**0.957**	**0.982**	**0.913**	**0.633**
nyfed_weekly	24	52	**0.150**	NA	**0.977**	**0.862**^{M01}	**0.211**	0.025	**0.925**	**0.979**	**0.862**	**0.926**	**0.977**	**0.925**
nyfed_weekly	24	260	0.000	0.000	0.066	**0.261**	**0.201**	0.000	**0.487**	0.072	**1.000**	**0.336**	**0.487**	0.066

Table S41: Treasury bills; method block 3/3. Full-grid $n = 277$.

Source	Horizon	History	Rolling peak	Rolling transl.	Mean combination	Median combination
nyfed_weekly	6	52	**1.000**	**0.516**	**0.940**	**0.809**^{M01}
nyfed_weekly	6	260	**1.000**	**0.663**	**0.622**	**0.848**
nyfed_weekly	24	52	**0.926**	**1.000**	**0.925**	**0.862**^{M01}
nyfed_weekly	24	260	**0.624**	0.066	**0.336**	**0.321**

Other NYFed panels. Each method column is shown once for every row of this group.

Table S42: Other NYFed panels; method block 1/1. Full-grid $n = 277$.

Source	Horizon	History	Hist. headroom	Intercept only	Finite ridge	Expanded ridge	One-SE ridge	Multi-fold one-SE	Expanded full reg.	Four-feature	Endpoint plug-in	Original full reg.	Mean combination	Median combination
PDPOSCS-TOT	6	52	**0.312**	**1.000**^{M02}	**0.312**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**1.000**^{M02}	**0.312**^{M14}	**0.312**	**0.172**	0.095	**0.312**	NA
PDPOSCS-TOT	6	260	**0.582**	**0.780**^{M02}	**0.243**^{M14}	**0.243**^{M14}	**0.780**^{M02}	**0.780**^{M02}	**0.243**^{M14}	**0.780**	0.000	**0.697**	**1.000**	NA
PDPOSCS-TOT	24	52	**0.898**	**0.898**^{M02}	**0.780**^{M14}	**0.898**^{M02}	**0.898**^{M02}	**0.898**^{M02}	**0.780**^{M14}	**0.898**	**1.000**	0.085	**0.780**	NA
PDPOSCS-TOT	24	260	0.000	**1.000**^{M02}	0.010^{M14}	**1.000**^{M02}	**1.000**^{M02}	**1.000**^{M02}	0.010^{M14}	0.000	0.000	0.003	0.000	NA
PDPOSFGS-TOT	6	52	**0.248**	**1.000**^{M02}	**0.868**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**1.000**^{M02}	**0.868**^{M14}	**0.968**	**0.974**	0.014	**0.492**	NA
PDPOSFGS-TOT	6	260	0.017	0.039	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	0.001	0.000	**0.709**	**0.709**	NA
PDPOSFGS-TOT	24	52	**0.521**	**0.390**^{M02}	**0.307**^{M14}	**0.307**^{M14}	**0.390**^{M02}	**0.390**^{M02}	**0.307**^{M14}	**0.521**	**1.000**	**0.133**	**0.390**	NA
PDPOSFGS-TOT	24	260	0.010	**0.772**	**0.772**^{M14}	**0.772**^{M14}	**0.772**^{M14}	**0.772**^{M14}	**0.772**^{M14}	0.000	**1.000**	**0.470**	**0.425**	NA
PDPOSGSC-G2L3	6	52	**1.000**	**0.208**	**0.189**^{M14}	**0.189**^{M14}	**0.189**^{M14}	**0.189**^{M14}	**0.189**^{M14}	0.019	0.035	0.018	**0.189**	NA
PDPOSGSC-G2L3	6	260	**0.676**	**0.676**	**0.118**^{M14}	**0.118**^{M14}	**0.676**^{M16}	**0.676**^{M16}	**0.118**^{M14}	**0.348**	**1.000**	**0.118**^{M14}	**0.521**	NA
PDPOSGSC-G2L3	24	52	**1.000**	**0.280**^{M02}	0.004^{M14}	0.004^{M14}	0.069	**0.280**^{M02}	0.004^{M14}	0.005	0.065	0.004^{M14}	0.069	NA
PDPOSGSC-G2L3	24	260	**0.204**	**0.204**^{M02}	**0.271**^{M14}	**0.204**^{M02}	**0.204**^{M02}	**0.204**^{M02}	**0.271**^{M14}	**0.204**	**1.000**	**0.245**	**0.271**	NA
PDPOSGSC-G3L6	6	52	**0.624**	**0.921**	**0.426**^{M14}	**0.426**^{M14}	**0.602**^{M16}	**0.602**^{M16}	**0.426**^{M14}	0.033	0.078	0.025	**1.000**	NA
PDPOSGSC-G3L6	6	260	**0.746**	**0.437**^{M02}	**0.710**^{M14}	**0.710**^{M14}	**0.437**^{M02}	**0.437**^{M02}	**0.710**^{M14}	**0.437**	**0.461**	**0.710**^{M14}	**1.000**	NA
PDPOSGSC-G3L6	24	52	**0.471**	**0.471**^{M02}	**0.460**^{M14}	**0.460**^{M14}	**0.460**^{M14}	**0.471**^{M02}	**0.460**^{M14}	0.039	**1.000**	0.009	0.039	NA
PDPOSGSC-G3L6	24	260	**0.103**	**0.127**	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**0.103**	**0.997**	**0.678**	**0.703**	NA
PDPOSGSC-G6L7	6	52	**0.962**	**1.000**^{M02}	**0.119**^{M14}	**0.119**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**0.119**^{M14}	**0.119**	**0.962**	0.000	**0.258**	NA
PDPOSGSC-G6L7	6	260	**0.366**	**0.366**	0.089^{M14}	0.089^{M14}	0.089^{M14}	0.089^{M14}	0.089^{M14}	0.089	**1.000**	0.089^{M14}	**0.364**	NA
PDPOSGSC-G6L7	24	52	**1.000**	**0.305**	0.000^{M14}	0.000^{M14}	0.000^{M14}	**0.356**	0.000^{M14}	0.000	**0.929**	0.000^{M14}	**0.356**	NA
PDPOSGSC-G6L7	24	260	**1.000**	**0.407**	**0.131**^{M14}	**0.131**^{M14}	**0.162**^{M16}	**0.162**^{M16}	**0.131**^{M14}	**0.192**	**0.407**	**0.131**^{M14}	**0.407**	NA
PDPOSGSC-G7L11	6	52	**1.000**	**0.166**^{M02}	**0.143**^{M14}	**0.143**^{M14}	**0.143**^{M14}	**0.166**^{M02}	**0.143**^{M14}	**0.166**	**0.807**	0.014	**0.166**	NA
PDPOSGSC-G7L11	6	260	**0.806**	**0.702**^{M02}	**0.609**^{M14}	**0.609**^{M14}	**0.702**^{M02}	**0.702**^{M02}	**0.609**^{M14}	**0.806**	**0.806**	**0.609**^{M14}	**1.000**	NA
PDPOSGSC-G7L11	24	52	**0.459**	**0.136**^{M02}	**0.459**^{M14}	**0.136**^{M02}	**0.136**^{M02}	**0.136**^{M02}	**0.459**^{M14}	0.092	**1.000**	**0.120**	**0.371**	NA
PDPOSGSC-G7L11	24	260	**1.000**	**0.488**	0.073^{M14}	0.073^{M14}	**0.253**^{M16}	**0.253**^{M16}	0.073^{M14}	0.078	**0.795**	0.073^{M14}	**0.253**	NA
PDPOSGSC-L2	6	52	**0.305**	**0.614**^{M02}	0.000^{M14}	0.000^{M14}	**0.614**^{M02}	**0.614**^{M02}	0.000^{M14}	**1.000**	0.002	0.000^{M14}	**0.130**	NA
PDPOSGSC-L2	6	260	**0.862**	**0.830**^{M02}	**0.862**^{M14}	**0.862**^{M14}	**0.830**^{M02}	**0.830**^{M02}	**0.862**^{M14}	**0.200**	**1.000**	**0.193**	**0.486**	NA
PDPOSGSC-L2	24	52	**1.000**	**0.202**^{M02}	**0.202**^{M14}	**0.202**^{M14}	**0.202**^{M14}	**0.202**^{M02}	**0.202**^{M14}	**0.209**	**0.928**	0.002	**0.174**	NA
PDPOSGSC-L2	24	260	**0.359**	**0.359**^{M02}	**0.359**^{M14}	**0.359**^{M02}	**0.359**^{M02}	**0.359**^{M02}	**0.359**^{M14}	**0.359**	**1.000**	**0.149**	**0.359**	NA
PDPOSMBS-TOT	6	52	**1.000**	**0.576**^{M02}	**0.464**^{M14}	**0.576**^{M02}	**0.576**^{M02}	**0.576**^{M02}	**0.464**^{M14}	**0.576**	**0.576**	0.000	**0.576**	NA
PDPOSMBS-TOT	6	260	**0.902**	**0.415**^{M02}	**0.192**^{M14}	**0.192**^{M14}	**0.415**^{M02}	**0.415**^{M02}	**0.192**^{M14}	**0.493**	**0.798**	**0.192**^{M14}	**1.000**	NA
PDPOSMBS-TOT	24	52	**0.110**	**0.110**^{M02}	0.072^{M14}	0.072^{M14}	0.072^{M14}	**0.110**^{M02}	0.072^{M14}	**0.110**	**1.000**	0.018	**0.110**	NA
PDPOSMBS-TOT	24	260	**0.784**	**0.784**^{M02}	**0.784**^{M14}	**0.784**^{M02}	**0.784**^{M02}	**0.784**^{M02}	**0.784**^{M14}	**0.784**	**1.000**	**0.784**	**0.784**	NA
PDPOSTIPS-G11	6	52	**0.832**	**1.000**	**0.282**^{M14}	**0.282**^{M14}	**0.282**^{M14}	**0.282**^{M14}	**0.282**^{M14}	**0.282**	**0.261**	0.000	**0.238**	NA
PDPOSTIPS-G11	6	260	0.011	**1.000**^{M02}	**0.106**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**1.000**^{M02}	**0.106**^{M14}	0.096	0.014	0.068	**0.135**	NA
PDPOSTIPS-G11	24	52	**0.673**	**0.673**^{M02}	0.037^{M14}	0.037^{M14}	**0.673**^{M02}	**0.673**^{M02}	0.037^{M14}	0.018	**1.000**	0.004	0.037	NA
PDPOSTIPS-G11	24	260	**0.449**	**1.000**	**0.449**^{M14}	**0.449**^{M14}	**0.449**^{M14}	**0.449**^{M14}	**0.449**^{M14}	**0.506**	0.012	**0.449**^{M14}	**0.506**	NA
PDPOSTIPS-G2	6	52	**1.000**	**0.482**^{M02}	**0.241**^{M14}	**0.241**^{M14}	**0.482**	**0.482**^{M02}	**0.241**^{M14}	**0.445**	**0.245**	0.004	**0.245**	NA
PDPOSTIPS-G2	6	260	**0.701**	**1.000**	0.057^{M14}	0.057^{M14}	0.057^{M14}	0.057^{M14}	0.057^{M14}	**0.581**	**0.215**	0.057^{M14}	0.001	NA
PDPOSTIPS-G2	24	52	**0.283**	**1.000**^{M02}	**0.169**^{M14}	**0.169**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**0.169**^{M14}	**0.426**	**0.255**	0.006	**0.169**	NA
PDPOSTIPS-G2	24	260	**1.000**	**0.663**^{M02}	0.078^{M14}	0.078^{M14}	**0.663**^{M02}	**0.663**^{M02}	0.078^{M14}	**0.200**	**0.701**	0.078^{M14}	**0.419**	NA
PDPOSTIPS-G6L11	6	52	**0.437**	0.000^{M02}	**0.437**^{M14}	0.000^{M02}	0.000^{M02}	0.000^{M02}	**0.437**^{M14}	**0.437**	**0.162**	0.025	**1.000**	NA
PDPOSTIPS-G6L11	6	260	0.018	**0.101**^{M02}	**0.101**^{M14}	**0.101**^{M02}	**0.101**^{M02}	**0.101**^{M02}	**0.101**^{M14}	**0.101**	0.000	**0.887**	**1.000**	NA
PDPOSTIPS-G6L11	24	52	**0.280**	0.001^{M02}	**0.231**^{M14}	**0.231**^{M14}	0.001^{M02}	0.001^{M02}	**0.231**^{M14}	**0.231**	**0.477**	0.071	**1.000**	NA
PDPOSTIPS-G6L11	24	260	0.000	0.000^{M02}	0.000^{M14}	0.000^{M02}	0.000^{M02}	0.000^{M02}	0.000^{M14}	0.000	0.000	**1.000**	0.000	NA
PDPOSTIPS-L2	6	52	**0.921**	**0.921**^{M02}	**0.921**^{M14}	**0.921**^{M02}	**0.921**^{M02}	**0.921**^{M02}	**0.921**^{M14}	**0.258**	**1.000**	0.000	0.075	NA
PDPOSTIPS-L2	6	260	**0.460**	**0.294**	0.047^{M14}	0.047^{M14}	**0.301**^{M16}	**0.301**^{M16}	0.047^{M14}	**0.460**	**1.000**	0.047^{M14}	**0.460**	NA
PDPOSTIPS-L2	24	52	**0.247**	**0.247**	0.015^{M14}	0.015^{M14}	0.015^{M14}	**0.247**	0.015^{M14}	**0.230**	**1.000**	0.000	0.015	NA
PDPOSTIPS-L2	24	260	**0.427**	**0.248**^{M02}	0.000^{M14}	0.000^{M14}	**0.248**^{M02}	**0.248**^{M02}	0.000^{M14}	0.012	**1.000**	0.000^{M14}	0.002	NA

Same-record dailyFX. Each method column is shown once for every row of this group.

Table S43: Same-record dailyFX; method block 1/1. Full-grid $n = 501$.

Source	Horizon	History	Hist. headroom	Full regression	Four-feature
fx_daily	6	52	**1.000**	0.005	0.005
fx_daily	6	260	**0.604**	0.087	**1.000**
fx_daily	24	52	**1.000**	0.011	0.041
fx_daily	24	260	**1.000**	**0.347**	**0.581**

CFTC. Each method column is shown once for every row of this group.

Table S44: CFTC; method block 1/1. Full-grid $n = 260$.

Source	Horizon	History	Hist. headroom	Intercept only	Cal. history	Cal. full regression	Full regression	Four-feature
cftc:020601	6	52	**0.774**	**0.309**	**0.133**	**1.000**	0.053	**0.133**
cftc:020601	6	260	**0.242**	**0.234**	**1.000**	**0.819**	**0.242**	**0.242**
cftc:042601	6	52	**1.000**	**0.540**	**0.264**	**0.430**	**0.264**	**0.540**
cftc:042601	6	260	**0.957**	**1.000**	**0.957**	**0.892**	**0.629**	0.068
cftc:043602	6	52	**1.000**	**0.258**[M08]	**0.101**	**0.101**	**0.258**[M08]	**0.126**
cftc:043602	6	260	**0.773**	**0.171**	**0.927**	**1.000**	**0.200**	**0.221**
cftc:044601	6	52	**1.000**	**0.323**	**0.323**	**0.196**	**0.196**	0.043
cftc:044601	6	260	**1.000**	**0.457**	**0.457**	0.092	0.069	**0.229**
cftc:045601	6	52	**0.810**	**1.000**[M08]	**0.120**	**0.810**	**1.000**[M08]	**0.810**
cftc:045601	6	260	**1.000**	**0.164**[M08]	**0.967**	**0.967**	**0.164**[M08]	**0.164**[M08]
cftc:090741	6	52	**1.000**	**0.578**[M08]	**0.578**	**0.578**	**0.578**[M08]	**0.160**
cftc:090741	6	260	**0.915**	**0.372**	**0.915**	**1.000**	**0.962**	**0.985**
cftc:092741	6	52	**0.986**	**0.986**[M08]	**1.000**	**0.978**	**0.986**[M08]	**0.986**[M08]
cftc:092741	6	260	**0.761**	**0.761**	**0.761**	**0.761**	**0.724**	**1.000**
cftc:095741	6	52	**1.000**	0.067	0.067	0.048	0.067	**0.142**
cftc:095741	6	260	**0.823**	**1.000**	**0.542**	0.016	**0.121**	**0.811**
cftc:096742	6	52	**1.000**	**0.876**	0.009	0.009	**0.876**	**0.876**
cftc:096742	6	260	**0.982**	**0.324**	**0.162**	**1.000**	0.059	**0.966**
cftc:097741	6	52	**0.790**	**1.000**	**0.263**	**0.368**	**0.263**	**0.368**
cftc:097741	6	260	**0.955**	**0.848**	**0.955**	**0.955**	**0.612**	**1.000**
cftc:098662	6	52	**0.986**	**1.000**[M09]	0.013	0.009	**0.986**	**1.000**[M09]
cftc:098662	6	260	0.016	0.020[M08]	0.011	0.011	0.020[M08]	**1.000**
cftc:099741	6	52	**0.624**	**0.624**	0.065	**1.000**	**0.510**	**0.624**
cftc:099741	6	260	**1.000**	**0.861**[M08]	0.095	0.095	**0.861**[M08]	**0.861**[M08]
cftc:112741	6	52	**1.000**	**0.178**	**0.131**	**0.131**	**0.178**	**0.178**
cftc:112741	6	260	**1.000**	**0.298**	**0.831**	**0.634**	0.037	**0.884**
cftc:1170E1	6	52	**0.981**	**0.582**	**1.000**	**0.582**	**0.182**	**0.582**
cftc:1170E1	6	260	0.000	0.000	**0.110**	**0.179**	**0.278**	**1.000**
cftc:232741	6	52	**1.000**	**0.959**	0.053	0.003	0.099	0.066
cftc:232741	6	260	**0.976**	**0.732**	**0.976**	**1.000**	**0.732**	**0.976**
cftc:240743	6	52	0.042	0.014	0.042	0.001	0.005	**1.000**
cftc:240743	6	260	0.000	**1.000**[M08]	0.019	0.036	**1.000**[M08]	**1.000**[M08]
cftc:33874A	6	52	**1.000**	**0.241**	**0.241**	**0.177**	**0.241**	0.014
cftc:33874A	6	260	0.003	**0.926**	**0.926**	**0.921**	**1.000**	**0.926**

Complete method legend.

F/I are canonical matched aliases within each comparison only; this does not equate forecasts across sources. Class codes: H=matched historical F/I; R=other historical (rolling F); C=conditional procedure; B=other benchmark; U=unclassified known NA. Conditional procedures retain class C even when they select infinity or share exact losses with F/I. Superscripts identify exact full-loss classes, not semantic method mergers. Known NA IDs retain their original full spelling.

Short ID	Unchanged full method ID	Class
M01	F	H
M02	I	H
M03	ACI_SAV::ACI_F	C
M04	ACI_SAV::ACI_QR	C
M05	ACI_SAV::SAV	C
M06	CFTC::ACI_F	C
M07	CFTC::ACI_QR33	C
M08	CFTC::QR33	C
M09	CFTC::QR4	C
M10	E3_extended_menu::QR33_plus	C
M11	Levy::Levy_L	B
M12	QR4::QR4	C
M13	R6::R3	C
M14	R6::R6_R0	C
M15	R6::R6_R1	C
M16	R6::R6_R2	C
M17	R6::R6_R3	C
M18	R6_original_structure::retro_original_3menu_QR33	C
M19	R6_original_structure::retro_primary_F_reference	R
M20	daily_FX::QR33	C
M21	daily_FX::QR4	C
M22	mature_baselines::FHS	C
M23	mature_baselines::PEAK_POT	B
M24	mature_baselines::QRF	C
M25	original::KNN	C
M26	original::QR	C
M27	original::SCALED_KNN	C
M28	original::frozen_K	B
M29	original::frozen_S	B
M30	original::one_state_F	C
M31	original::one_state_K	C
M32	original::one_state_S	C
M33	original::rolling_F	R
M34	original::rolling_K	B
M35	original::rolling_S	B
M36	panel_E5::QR33_plus	C
M37	panel_legacy::C2_QR4	C
M38	panel_legacy::Levy_L	B
M39	panel_legacy::QR33	C
M40	published_combination::MEAN_F_QR	C
M41	published_combination::MEDIAN_F_QR_KNN	C

The display is exhaustive but descriptive across related, overlapping tasks; it does not pool independent-source p-values or establish future risk coverage. The machine display map preserves full-precision entries, NA metadata, every class representative and per-comparison storage aliases.

S32 Fixed limit costs and full parameter regions

For each comparison, let B be the issue-known floor, Y the realized sampled peak, C_m a saved method report, and $m_F = \overline{C_F - B} > 0$ the matched F report's mean reserved headroom. All four quantities use the same source-native position units (currency or contract counts). The normalized per-issue cost is

$$J_m(\kappa, \beta, c) = \kappa \frac{\overline{C_m - B}}{m_F} + \beta\, \overline{\mathbf{1}\{Y > C_m\}} + c \frac{\overline{(Y - C_m)_+}}{m_F}, \qquad b = \beta m_F.$$

Reserve, strict breach frequency and excess enter separately. The fixed breach charge b is in those same position-value units; each β scales it to that comparison's evaluation-sample F headroom. Charges therefore differ across comparisons. Costs apply per issued report; overlapping horizons are not added as separate cash flows. The fixed grid is $\kappa \in \{.01, .025, .05, .1, .2, .5, 1\}$, $\beta \in \{0, .01, .05, .1, .25, .5, 1, 2, 5\}$ and $c \in \{.5, 1, 2\}$. At $\kappa = .05, \beta = 0, c = 1$, the cost difference from F is the pinball-loss difference divided by m_F. Other weights need not preserve that ranking. The panels compare fixed reports and do not estimate a cost-optimal policy.

All 98 comparisons have a positive scale and all 976 complete methods enter 189 grid points, giving 184,464 cost rows. Sixteen missing-floor inputs are recovered only from their registered, fingerprinted same-origin F copies. Their complete origin, target and floor arrays match exactly. Other missing or mismatched floors would make the whole comparison unavailable. No missing scale uses an epsilon.

The public `results/v10/cost/cost-grid.csv.zip` contains every complete method's normalized mean cost, its F reference and strict cheaper/equal status at all coefficients. `primary-regions.csv` retains all 18,522 planned primary grid rows, including status fields. Primitive reserve amounts, strict breach counts/rates and excess amounts appear in `primitive-stats.json`. Source-group figures use the full planned comparison denominators. The primary comparison is fixed expanded-menu full regression for hourly/weekly/CFTC sources and the original saved regression for daily FX. Other methods retain separate regions and are never replaced by a minimum-cost oracle.

S33 Retained accuracy and hit diagnostics

These are the unchanged earlier tests and Murphy plots moved from the main article. The p-values retain their original named adjustment families; no MCS adjustment is retroactively assigned to them.

Table S46: Accuracy and hit diagnostics for explicitly named reports. Panel A reports paired working-reference tests; Panel B uses the fixed nonoverlap grid for hit tests. A better average score does not imply fewer breaches.

Panel A: paired contrast	DM p	HLN p	Named adjustment / p
FX original main comparison	2.20e-07	2.31e-07	Two-main comparison Bonferroni: 4.39e-07
FX matched history	4.47e-09	4.83e-09	Original matched Holm family: 2.86e-07
Treasury original main comparison	0.00761	0.00939	Original matched Holm family: 0.320

Original matched Holm uses 72 contrasts. The later regrouped matched family is separate; earlier-period writing-stage HAC comparisons have neither a new HLN field nor a new family adjustment.

Panel B: report	Grid hits	Grid issues	Kupiec p	Christoffersen CC p
FX regression	50	947	0.695	0.636
FX original rolling reference	49	947	0.807	0.905
FX matched fixed reference	49	947	0.807	0.646
Treasury regression	7	40	0.00413	0.0136
Treasury historical reference	4	40	0.199	0.262

Hit tests use the full fixed nonoverlap grid; the conditional-coverage likelihood conditions on its initial hit. Equality to the nominal tail rate is the stated diagnostic null, distinct from the population upper-quantile inequality. The grid does not create independence.

S34 Reproduction, availability and retained extended results

Audit and run records are retained verbatim in the v10b audit archive.

Editable sources, aggregate results, original designs and code accompany the paper; favourable, adverse and failed outputs remain accessible.

Document regeneration checks saved aggregates without raw observations; it does not establish recomputed forecast accuracy or independent replication.

Full financial replay requires compatible inputs. A separately authorized replay requires the original FX member and Treasury snapshot to match their recorded fingerprints and definitions. Restricted FX observations and private per-origin forecasts or fitted traces are not redistributed. Downloaded economic dates are not verified first-release availability; a matching series name does not establish snapshot identity.

Routes to complete evidence. The v6 replay guide, original conditional archive, supervisor rule archive and extended-results index identify the complete retained tables, failure inventories and longer technical discussions. Individual-dealer external validation and prospective first-release validation remain unmet.

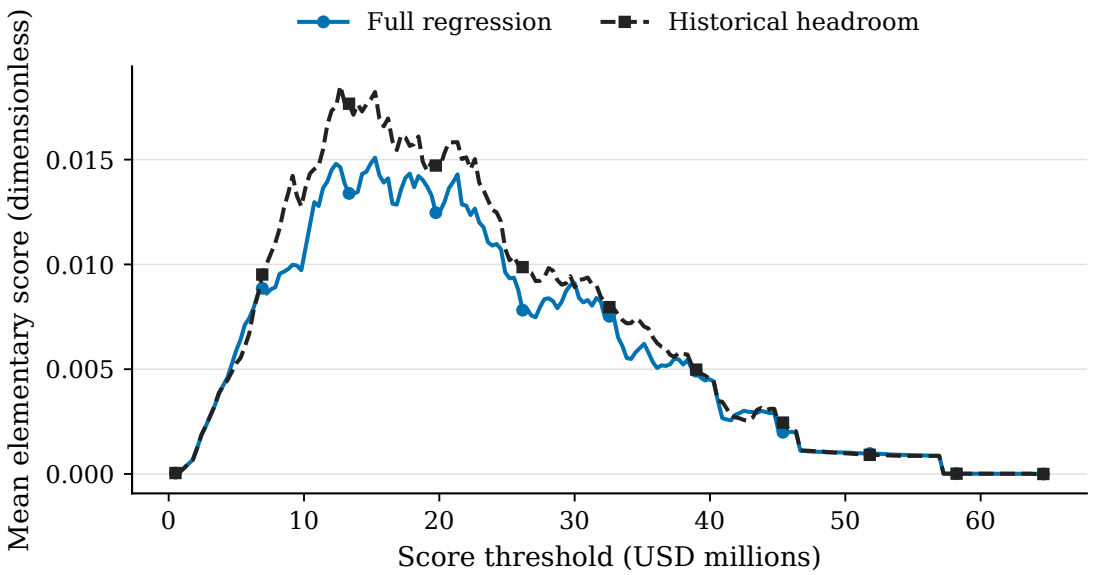


Figure S1: Murphy diagram for the main short-horizon FX comparison. The full-breakpoint calculation finds crossings; the graph is a display of saved scores. Average pinball superiority does not imply dominance across elementary scores.

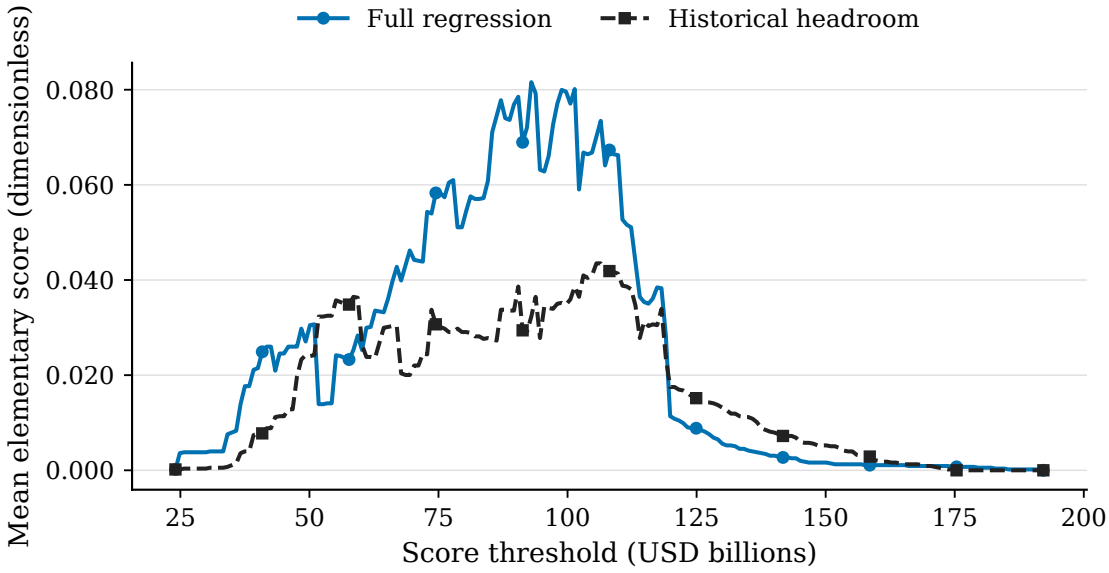


Figure S2: Murphy diagram for the main Treasury-bill comparison. The saved full-breakpoint comparison also crosses. A single average-score ordering leaves other consistent-score preferences unresolved.